

RCF v2.0

Structural Rewrite

Change Log

A deep-dive comparison of version 2.0 against version 1.0 of the Reconciliation Causal Framework. Covers section remapping (10 → 8 sections, -63.5% by character count), conceptual reframings (strip-and-defer acyclicity, Three-Tier Burden, QM-layer consolidation), per-section change analysis with verbatim before/after quotes, and the four open theorem targets carried forward into Phase C.

10 → 8
SECTIONS

131
DISTINCT
CHANGES

9
HONEST
DOWNGRADES

4
OPEN TARGETS

AUDIENCE
DETAIL LEVEL
DATE

Internal Framework Team
Deep Dive · Before/After Quotes
2026-07-31

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Chapter 1 — Executive Summary

This document is a deep-dive change log comparing version 2.0 of the **Reconciliation Causal Framework (RCF)** against version 1.0. The RCF is a structural attempt to derive physics — algebraic foundation, causality, spacetime geometry, matter, gravity, black holes, probability, and cosmology — from a single relational algebra equipped with a master constraint and a fracture operator. v1.0 was delivered as ten merged canonical-form PDFs (Sections 0–9, ~777K characters total); v2.0 is delivered as eight rewritten canonical-form PDFs (Sections 0–7, ~283K characters total). v2.0 is **63.5% shorter** than v1.0, but the size reduction understates the depth of the rewrite: v2.0 is a structural reorganization, not a compression.

The headline numbers: ten sections collapse to eight; 131 distinct changes were identified across the eight v2.0 sections; 6 explicit circularity admissions and 9 status downgrades replace v1.0's overclaims; 2 major terminology renames (SOE/MOE → Open Expansion/Open Extension) propagate through every section; and 1 entirely new physical mechanism (the dark energy mechanism at §2.2) is introduced. 4 of the seven theorem targets (T-1, T-2, T-5, T-7) remain open, with T-2 explicitly identified as the existential central open problem.

Key Restructuring at a Glance

v1.0 §0 (Algebraic Foundation) was overloaded: it contained the Reconciliation Propagator, the Convergence Theorem, the thin/full A_{phy} split, the Dirac bracket, and the Reconciliation Principle — all of which depend on conceptual ingredients (the fracture, burden, Tier 1/Tier 2 split, correlation kernel) that v1.0 had not yet introduced. v2.0 strips §0 to its foundational core and distributes the displaced content to §1 (the new home for the fracture machinery and propagator) and §2 (the new home for the Reconciliation Principle and the QM layer). v1.0 §3 (Time) and §5 (Gravity) are merged into v2.0 §4 because they belong to the same GR-perspective emergence layer. v1.0 §4 (Matter) and §7 (Probability) are consolidated with the Reconciliation Principle into v2.0 §2 as the QM layer.

The rewrite is driven by two organizing principles. The first is **acyclicity as a design principle**: v1.0 had patched circularities with forward-reference contracts and a $\delta\hat{C}$ replacement; v2.0 eliminates the circularities structurally by moving every object to the section where its conceptual ingredients first exist. Each v2.0 section closes with an explicit Acyclicity Test that itemizes every object's dependencies and issues a verdict. The second is an **honesty campaign**: v1.0 had overclaimed numerous results as "Established" or "Strengthened" when their proofs rested on unproved premises or circular arguments; v2.0 either fixes the proofs or honestly downgrades the status to "Conditional," "Conjecture," or "Principle."

The remainder of this document is organized as follows. Chapter 2 presents the section-mapping table and size data. Chapter 3 identifies seven cross-cutting themes that span multiple sections. Chapter 4 — the bulk of the document — provides a deep-dive per-section analysis, with before/after quotes and impact assessment for each significant change. Chapter 5 assesses the aggregate impact of the rewrite on the framework as a whole. Chapter 6 lists the open issues carried forward into Phase C.

Chapter 2 — Restructuring Overview

v2.0 collapses ten sections into eight by three structural moves: (a) consolidating QM-layer objects (Reconciliation Principle, fields, particles, mass, records, Born rule, FIREWALL) into a new §2; (b) merging Time + Gravity into a new §4; and (c) splitting v1.0 §0's prematurely-placed content into a narrowed §0 (algebraic foundation only) and a reborn §1 (fracture machinery, propagator, and convergence). The table below maps each v1.0 section to its v2.0 counterpart and identifies the change type.

2.1 Section Mapping (v1.0 → v2.0)

v1.0 Source	v1.0 Title	v2.0 Target	v2.0 Title	Change Type
§0	Algebraic Foundation — Reconciliation Propagator	§0 + parts moved to §1	Algebraic Foundation and the Two Primitives	STRIPPED — content narrowed; R_t, Convergence, A_red, M_red, RP moved out
§1	Causal Foundation — Zero-Preserving Skeleton	§1	Open Expansion, Fracture, and the Reconciliation Propagator	REBORN — absorbs displaced v1 §0 content + Open Extension from v1 §1
§2	Emergent Space — Correlation Geometry	§3	Emergent Geometry	RENUMBERED — largely intact, 4 targeted corrections, Π_{net} relocated in
§3 + §5	Emergent Time + Gravity	§4	Time, Gravity, and the Three-Tier Burden	MERGED — 2 v1 sections into 1, 5 corrections, Three-Tier reframing
§4 + §0.8 + §7	Fields & Particles + RP + Probability	§2	Burden, Matter, and the Reconciliation Principle	CONSOLIDATED — 4 v1 sections into 1 (the QM layer)
§6	Black Holes — Unreconciled Relational Sectors	§5	Black Holes	RENUMBERED — largely intact, 6 corrections, layer-badge fixed
§8	Cosmology — The Open Universe	§6	Cosmology	RENUMBERED — largely intact, 7 corrections, dark energy mechanism linked to §2.2
§9	Audit — Manuscript Closure	§7	Audit, Formal Action, and Closure	RENUMBERED — largely intact, 6 corrections, theorem-status table rebuilt

2.2 Size Data (Characters per Section)

The following table shows the character count per section in v1.0 and v2.0. Note that v1.0 §0, §1, §2, §3, §4, §5, §6, §7, §8, §9 contained 64K, 72K, 72K, 63K, 86K, 81K, 88K, 74K, 75K, and 87K characters respectively. v2.0 §0–§7 contained 45K, 41K, 41K, 31K, 32K, 24K, 25K, and 38K characters respectively. The reductions are overwhelmingly apparatus loss (patch scaffolding, source citations, epistemic tags, redundant summary tables, prose essays), not content loss.

v2.0 Section	v2.0 Title	v2.0 Chars	v1.0 Source(s)	v1.0 Chars	Delta
§0	Algebraic Foundation and the Two Primiti...	44,786	§0	64,105	-19,319 (-30.1%)
§1	Open Expansion, Fracture, and the Reconc...	40,625	§0	64,105	-23,480 (-36.6%)
§2	Burden, Matter, and the Reconciliation P...	40,625	§0	64,105	-23,480 (-36.6%)
§3	Emergent Geometry	30,527	§2	71,927	-41,400 (-57.6%)
§4	Time, Gravity, and the Three-Tier Burden	32,263	§3	63,290	-31,027 (-49.0%)
§5	Black Holes	23,577	§6	87,958	-64,381 (-73.2%)
§6	Cosmology	25,063	§8	74,556	-49,493 (-66.4%)
§7	Audit, Formal Action, and Closure	38,155	§9	86,819	-48,664 (-56.1%)

Aggregate Size Delta

v1.0 total: 776,954 characters across 10 sections. v2.0 total: 283,430 characters across 8 sections. Net delta: -493,524 characters (-63.5%). The reduction comes overwhelmingly from apparatus (patch scaffolding P1–P7, source citations, epistemic tags, redundant summary tables, prose essays); the substantive content survives intact.

Chapter 3 — Cross-Cutting Themes

Seven themes recur across multiple sections of v2.0. Understanding these themes up front makes the per-section analysis in Chapter 4 easier to follow, because the same conceptual move appears in different guises in different sections. The themes are: (1) the strip-and-defer strategy that achieves acyclicity; (2) the honesty campaign of downgrades and circularity fixes; (3) the SOE/MOE → Open Expansion/Open Extension terminology rename; (4) the forward-reference cleanup; (5) the new dark energy mechanism at §2.2; (6) the M_{red} gap closure; and (7) the layer-scheme reconciliation that retires the A/B/C Three-Layer Protocol in favor of $L \rightarrow Q \rightarrow C \rightarrow Q'$.

3.1 The Strip-and-Defer Strategy (Acyclicity as Design Principle)

The single organizing move of v2.0 is what we may call "strip and defer." v1.0 §0 had named the $L \rightarrow Q \rightarrow C \rightarrow Q'$ emergence ladder as its central architectural commitment, but in practice it had placed objects at §0 that depended on ingredients not yet available: the Reconciliation Propagator R_t at §0.4 required the fracture machinery (which v1.0 deferred to a later corollary of RP); the Reconciliation Principle at §0.8 required both causal order $<$ (forward-referenced from §1) and the correlation kernel K_ω (forward-referenced from §2); the cross-extension network operator Π_{net} at §0.3.8 forward-referenced K_ω from §2.1. v1.0 patched these circularities with a "forward-reference contract" box and the $\delta\hat{C}$ replacement for the §0.4 → §1.1.1 loop, but the patches left the dependency graph fragile.

v2.0 makes acyclicity a design principle rather than a defensive patch. Every object that depends on the fracture, on burden as a physical quantity, on R_t , on A_{red} , on M_{red} , on the Dirac bracket, or on the Reconciliation Principle is moved out of §0 into §1 or §2, where its conceptual ingredients exist. Each section then closes with an explicit Acyclicity Test that itemizes every object's dependencies by section number and issues a verdict. The verdicts are uniform: §0 acyclic, §1 acyclic, §2 acyclic (with one flagged forward reference to §3 for K_ω), §3 acyclic, §4 acyclic (with one flagged forward reference to §5 for Π_{max}), §5 acyclic, §6 acyclic, §7 acyclic. The cumulative dependency graph is promoted in §7.5 from v1.0's narrated assertion to a formally stated Theorem 7.5.1 with an itemized verification table.

3.2 The Honesty Campaign (Downgrades and Circularity Fixes)

v2.0 is, in part, a sustained honesty campaign. Across the eight sections, we count nine distinct downgrades of v1.0 status claims, plus six explicit circularity admissions. The downgrades follow a consistent pattern: v1.0 had claimed a result was "Established" or "Strengthened" when its proof actually rested on an unproved premise or a circular argument; v2.0 either fixes the proof or honestly downgrades the status to "Conditional," "Conjecture," or "Principle" (the latter reserved for variational assumptions that are not derivations).

The most visible downgrades: the $D=3$ closure "Theorem" (v1 §2.7.3) is demoted to "[Conditional Proposition]" (v2 §3.6.4) because the closure defect $\Xi_C(D)$ had been defined tautologically as $|D-3|$. The Einstein closure "Strengthened" (v1 §5.4.1 P5) is reverted to "Conditional" because the smooth 4D Lorentz-compatible continuum-limit premise (T-2) is open. The Bekenstein-Hawking "T-6 partially closed" (v1 §6.5.1) is downgraded to "T-6 area-scaling only; coefficient open" because v1.0 had assumed the exponential microstate growth that was its conclusion. Theorem 6.6 (Lowest-Burden Emission) and Theorem

6.7 (Hawking-Like Emission) in v1.0 §6 are both demoted — to Principle 5.6.1 and Conjecture 5.6.2 respectively — because their proofs "restated the conclusion." The $\Lambda_B = 0$ "Theorem 8.5.1" is downgraded to "Corollary 6.4.1" because it was conditional on T-2 and a theorem conditional on an unproved conjecture is not a theorem. The Q-9 "CLOSED (conditional on T-2)" status is reworded to "CONDITIONAL (on T-2)" because "CLOSED" and "conditional on an open target" cannot both hold.

The circularity admissions are equally candid. v1.0's Theorem 7.4 (sectorwise zero-decomposition) was content-free: it introduced $\mathcal{M}_B = \sum p_i \hat{C}_i^\dagger \hat{C}_i$ and noted $\omega(\mathcal{M}_B) = 0$, but the p_i were free parameters and the equation was satisfied trivially. v2.0 §2.7.1 restates the theorem using M_{red} (now properly defined at §1.4.3) so that positivity of M_{red} plus decoherence suppression actually constrains the p_i . v1.0's Theorem 5.1 (Symmetry of Burden Tensor) was tautological — "if Π is symmetric then Θ is symmetric" — and v2.0 derives $\Theta_{\mu\nu} = \Theta_{\nu\mu}$ from Hessian symmetry. v1.0's Theorem 4.1 (Particle Identity) had its hypotheses as its conclusion; v2.0 explicitly labels it a definitional equivalence. v1.0's Thm 5.2 (Active-Source Conservation) assumed the field equation it was supposed to derive; v2.0 derives conservation independently via diffeomorphism invariance of B_Δ .

3.3 Terminology Renamed: SOE/MOE → Open Expansion/Open Extension

v2.0 renames the two dynamical mechanisms that v1.0 had called SOE (Single Open Extension) and MOE (Multiple Open Extension). The new names are Open Expansion (for SOE) and Open Extension (for MOE). The rename is more than cosmetic. The v1.0 "Single/Multiple" framing implied discrete, isolated extension events; v2.0's "Open Expansion/Open Extension" framing captures the actual picture: local Jordan-built sectors continuously expand their algebraic scope (Open Expansion) to reconcile with the single global causal Lie layer (Open Extension). The mathematics is unchanged — the three-mechanism propagator $R_t = \lim_{N \rightarrow \infty} (R_{\{T/N\}^{\text{OExt}}} \circ (R_{\varepsilon^{\text{OExp}}}^N))$ has the same form as v1.0's $R_t = \lim_{N \rightarrow \infty} (R_{\{T/N\}^{\text{MOE}}} \circ (R_{\varepsilon^{\text{SOE}}}^N))$ — but the physical interpretation is sharpened.

The Definition 1.5.2/1.5.3 headers carry the parenthetical "(was SOE)" / "(was MOE)" for traceability, and the architectural summary table in §1.9 lists both names side-by-side. The rename propagates through every section that referenced SOE/MOE: §1's Local Convergence / Global Desynchronization theorem is now stated in Open Expansion/Open Extension terms; §3's phase-structure protection is rerouted to §1.5.2 Open Expansion spectral flow; §4's Arrow of Time generator assignment is corrected (Open Expansion = Hamiltonian, Open Extension = Lindblad — the opposite of v1.0's swapped assignment); §6's Friedmann equation is now "Open Extension descent on FLRW" rather than "MOE descent on FLRW." The Conceptual Integrity Firewall and Mechanism Assignment Summary tables in v1.0 §9 — which had carried an SOE/MOE/dephasing column for every theorem — are retired in v2.0 §7 in favor of descriptive mechanism phrases.

3.4 The Forward-Reference Cleanup

v1.0 carried an elaborate system of forward-reference contracts, RESOLVED boxes, and CERTIFIED callouts. The §0.8 Reconciliation Principle forward-referenced causal order $<$ from §1 and K_ω from §2. The §0.3.8 Π_{net} forward-referenced K_ω from §2.1. The §3.2.5 gravitational-time-dilation forward-referenced the mass-burden identity from §4.2.8. The §4.1.3 burden flux $J_{[ij]}(B)$ forward-referenced §3.1.4 (which did

not actually contain the object — a phantom reference). The §5.3.3 metric boundedness forward-referenced $\Pi_{\max}$ from §6.1. The §6.5 Hawking-Like Emission forward-referenced the Born rule from §7. v1.0 marked each forward reference with a boxed "Forward reference from §X — RESOLVED" callout and tracked them in §9's dependency graph.

v2.0 closes these forward references structurally rather than by contract. The §0.8 → §2.1 K_{ω} reference is closed by Def 2.1.2 (algebraic correlation strength $s(a,b)$ defined locally). The §0.3.8 → §2.1 K_{ω} reference is closed by relocating Π_{net} to §3.9 (after K_{ω} is available). The §3.2.5 → §4.2.8 mass-burden reference is closed by Cor 2.5.2 (with the corrected $\alpha(B_0) = 1/(1+\lambda \cdot m^2)$). The §4.1.3 → §3.1.4 phantom reference is closed by Def 2.3.3 (locally defined burden flux). The §5.3.3 → §6.1 $\Pi_{\max}$ reference is closed by Def 5.1.3 ($\Pi_{\max} = C_{\Pi}/\ell_0^2$) — though the coefficient C_{Π} remains open (T-2). The §6.5 → §7 Born rule reference is closed by Thm 2.7.4 (Born rule established in §2.7, no longer in §7). The boxed RESOLVED callouts disappear; in their place, each section's Acyclicity Test verifies the remaining forward references (mostly one-way dependencies on later sections, not circularities) and §7.5 Theorem 7.5.1 formally verifies the cumulative graph is acyclic.

3.5 The New Dark Energy Mechanism (§2.2)

The most consequential theoretical addition in v2.0 is the dark energy mechanism at §2.2 — entirely new physics with no v1.0 antecedent for the quantitative form. v1.0 §8.3.3 had treated dark energy only qualitatively as "the dynamic pressure of open extension at the causal frontier," with $\Lambda_B = 0$ EXACT (Thm 5.5) and the 120-order problem dismissed as a category error. v2.0 §2.2 keeps $\Lambda_B = 0$ and the category-error framing, but adds a positive mechanism: restorative power $\mathcal{P}_{\text{restore}} = \eta \cdot \|\nabla_{\{\text{Bures}\}} B_{\Delta}[\rho_t]\|^2$ (Def 2.2.1), burden-dependent relaxation rate $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$ (Prop 2.2.2), and dark-energy density $\rho_{\text{DE}} \sim (H/\Gamma)^2 \cdot \rho_{\text{source}}$ (Conj 2.2.3).

The 120-order suppression falls out as $(H_0 \cdot t_P)^2 \sim 10^{-122}$ because the exponent 2 is NOT tuned — it comes from burden being quadratic in mismatch ($B_{\Delta} = \|\text{mismatch}\|^2$). The mechanism predicts Γ was larger in the past (because B_{Δ} was larger in the past), so $\rho_{\text{DE}} \sim (1+z)^{-3}$ and $\Omega_{\text{DE}} \sim (1+z)^{-6}$, giving $\Omega_{\text{DE}} \sim 10^{-18}$ at recombination — eighteen orders below the CMB bound ($\Omega_{\text{EDE}} < 0.004$) — resolving the early-dark-energy tension. The cosmological application is deferred to §6, where Conjecture 6.3.2 imports the $(H/\Gamma)^2$ scaling, Theorem 6.3.3 derives the 120-order resolution, Theorem 6.3.4 derives the $(1+z)^{-6}$ suppression, Theorem 6.3.5 derives an evolving equation of state $w(t)$ with three falsifiable predictions against DESI/Euclid ($w > -1$ always; w evolves from 0 to -1 ; $\rho_{\text{DE}} \rightarrow 0$ as $t \rightarrow \infty$), and Corollary 6.3.6 establishes the asymptotic decay (upgraded from v1.0's Conjecture 8.3.3 because it follows directly from the Convergence Theorem).

3.6 The M_{red} Gap Closure

v1.0 had a quiet but consequential gap: the reduced master constraint M_{red} was referenced by §4.1.2 (field admissibility), §4.2.7 (Particle Identity), and §9.2's T-2 row ("shared hypothesis of Thm 4.2.2 and Thm 0.6.3"), but M_{red} was never formally defined anywhere in v1.0. The v1.0 §0.7b Dirac bracket subsection used A_{red} with $[\cdot, \cdot]_{\text{D}}$ but did not define M_{red} as an operator on A_{red} . Worse, the two theorems v1.0 §9.2 cited as consumers of M_{red} — "Thm 4.2.2" and "Thm 0.6.3" — did not actually exist (§4.2.2 was a subsection, not a theorem; Thm 0.6.3 was renumbered to 0.7.3, which is the thin=full equivalence).

v2.0 closes this gap in three moves. First, §1.4.3 introduces the formal definition: $M_{\text{red}} = \sum_{i \in I_2} w_i \cdot D_i^\dagger D_i \in A_{\text{red}}$, framed as "the SEPARATION of M into its Tier 1 and Tier 2 sub-collectives," not a "reduction." Remark 1.4.4 ties $\ker(M_{\text{red}})$ to the field admissibility condition in §2. Second, §2.7.1 rewrites the sectorwise zero-decomposition theorem (v1.0's content-free Thm 7.4) using $M_{\text{red}} = \oplus_k M_{\text{red}}^{\{(k)\}} + M_{\text{red}}^{\{\text{cross}\}}$; positivity of M_{red} plus decoherence suppression forces $\omega_k(M_{\text{red}}^{\{(k)\}}) = 0$ for each k with $p_k > 0$, constraining the p_k . Third, §7.2's T-2 row now references Theorem 1.7.4 (thin=full) and Theorem 2.5.1 (mass-burden identity) — the actual consumers — instead of the nonexistent v1.0 theorems. The downstream effect: the Born rule (Thm 2.7.4) now has a genuine objectivity premise (from M_{red} 's sectoral structure) rather than the vacuous premise it inherited from v1.0's tautological Thm 7.4.

3.7 Layer-Scheme Reconciliation: A/B/C Retired in Favor of $L \rightarrow Q \rightarrow C \rightarrow Q'$

v1.0 had two parallel layer schemes running simultaneously: the $L \rightarrow Q \rightarrow C \rightarrow Q'$ emergence ladder (Linear $\rightarrow$ Quadratic $\rightarrow$ Cubic $\rightarrow$ Quartic) used in §0 to describe the algebraic-to-geometric emergence, and the A/B/C Three-Layer Protocol used in §7–§9 to describe the quantum/probabilistic (A), exact-sector-metric (B), and coarse-grained-GR (C) layers. The two schemes did not align cleanly: v1.0 Layer A sat above L, Layer B sat somewhere between Q and C, and Layer C was both the emergence-ladder Cubic layer AND the coarse-grained GR layer. This created persistent confusion in §9's Layer Assignment Audit table, where the same object could be tagged "Layer B" in one column and "Quartic" in another.

v2.0 retires the A/B/C scheme entirely. The framework uses only $L \rightarrow Q \rightarrow C \rightarrow Q'$. v1.0 Layer A (full kinematic state, quantum-universal) is subsumed into the Cubic layer C (Sections 1–3, microscopic structure). v1.0 Layer B (exact sector metric) is also subsumed into C. v1.0 Layer C (coarse-grained GR geometry) is renamed Quartic layer Q' (Sections 4–6, coarse-grained hydrodynamics). Black holes (v1.0 §6, badged "Layer C exclusive") are rebadged Layer Q' in v2.0 §5, consistent with their actual content as coarse-grained hydrodynamic phenomena. Cosmology (v1.0 §8, badged "Layers B $\rightarrow$ C with A quarantined") is rebadged Layer Q' in v2.0 §6. §7.5 Remark 7.5.3 explicitly states: "The v1.0 'Three-Layer Protocol' (A/B/C) is retired. The framework uses only $L \rightarrow Q \rightarrow C \rightarrow Q'$." The FIREWALL guardrail (which v1.0 had phrased as "Layer A vs Layers B/C") is restated in §2.8.1 as "probabilistic (Layer A) vs algebraic (Layers A/B/C)" — and then, in the new vocabulary, as "probabilistic vs algebraic," with the layer tags dropped in favor of the actual distinction (linear functional ω vs operator algebra).

Chapter 4 — Section-by-Section Changes

This chapter provides a deep-dive analysis of each of the eight v2.0 sections. For each section we present: the mapping from v1.0 sources; the structural changes at a high level; the top changes (each with type, v1.0/v2.0 locations, verbatim before/after quotes, and an impact assessment); and a section-level impact assessment. Verbatim quotes preserve v1.0 and v2.0 mathematical unicode notation exactly ($\mathbb{F}$, $\mathbb{M}$, A_{phy} , R_t , Π_{net} , ℓ_0 , $\delta\hat{C}$, α_B , B_Δ , K_ω , K_ω , $\Theta^{(B)}_{\mu\nu}$, $G_{\mu\nu}$, Λ_B , κ_B , ρ_{DE} , Γ_{eff} , etc.).

4.1 v2.0 §0 — Algebraic Foundation and the Two Primitives

Change type: STRIPPED | Changes identified: 16 | v2.0 size: 44,786 chars

Mapping

v2.0 §0 is a NARROWED version of v1.0 §0. The v1.0 section carried nine substantive subsections (§0.0–§0.9 plus §0.7b) covering Layers L, Q, C, and Q' of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder. v2.0 §0 collapses this to four substantive subsections (§0.1–§0.4 plus §0.0 and §0.5) and reaches only Layers L and Q. Six entire v1.0 subsections — §0.4 (Reconciliation Propagator), §0.5 (Convergence Theorem), §0.6 (Thin A_{phy}), §0.7 (Full A_{phy} + Thm 0.7.3), §0.7b (Dirac Bracket), and §0.8 (Reconciliation Principle) — leave §0 entirely. Five move to v2.0 §1 (because they depend on the fracture machinery that does not emerge until §1); the Reconciliation Principle moves further, to v2.0 §2, because its variational target $I(S)$ requires the correlation strength $s(a,b)$ that is only available in §3.

Structural Changes

The v1.0 $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder — explicitly named as the central architectural commitment of v1.0 §0.0 — is dismantled as the organizing principle of §0. The Cubic (C) and Quartic (Q') layers no longer appear inside §0 at all. The badge "9 Subsections · 4 Layers · 1 Restored Theorem" disappears entirely. What stays is the genuinely foundational core: the unital complex (*)-algebra A (Def 0.1.1), primitive constraints, the master constraint M , the algebraic state, the constraint ideal, the compatibility bracket, the GNS construction, and an algebraic-only definition of the fracture operator F .

What is NEW — and is the single most consequential addition — is the Lie–Jordan decomposition (Def 0.1.2), used to ground two named co-primitives: causality (Lie algebra A_L) and locality (Jordan algebra A_J), introduced in v2.0 §0.4. v1.0 §0 contained no Lie–Jordan decomposition and no named causality/locality primitives; in v1.0 those objects lived in §1.1 (causal order) and were never explicitly tied to the Jordan structure. v2.0 turns the strip-and-defer policy into an explicit test: "does it contain anything that depends on fracture, sub-Hilbert spaces, burden, R_t , A_{red} , M_{red} , the Dirac bracket, or the Reconciliation Principle? If yes, the reordering has failed."

Top Changes (with Before/After Quotes)

► Section title renamed from "Reconciliation Propagator" to "Two Primitives"

Type: Renamed | v1.0: Title "Algebraic Foundation — Reconciliation Propagator" | v2.0: Title "Algebraic Foundation and the Two Primitives"

v1.0: "Section 0 / Algebraic Foundation / Reconciliation Propagator"

v2.0: "RCF Section 0 — Algebraic Foundation and the Two Primitives"

Impact. The title's load-bearing noun changes from "Reconciliation Propagator" (v1.0's central object, R_t) to "Two Primitives" (causality + locality). This signals the inversion of the section's center of gravity: in v1.0, §0 was the section that built the dynamical engine; in v2.0, §0 is the section that names the structural primitives and hands the dynamics to §1.

► $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder removed as §0 organizing principle

Type: Removed | v1.0: §0.0 (second paragraph) and the layer badges "LAYER $Q \cdot QUADRATIC$ ", "LAYER $Q' \cdot QUARTIC$ " | v2.0: not present; replaced by the "two-primitive picture" as the central architectural commitment

v1.0: "The central architectural commitment of this section is the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder."

v2.0: "The central architectural commitment of this section is the two-primitive picture, grounded in the Lie-Jordan decomposition of the algebra."

Impact. v1.0's four-layer emergence ladder was the spine of §0. v2.0 §0 only reaches Layers L and Q; C and Q' are no longer §0 concepts. The layer-badge callouts that prefixed every v1.0 subsection disappear from v2.0 §0 except for the two L/Q badges retained at §0.1 and §0.3.

► NEW — Lie–Jordan decomposition introduced at Layer L (Def 0.1.2)

Type: Added | v1.0: not present in v1.0 §0.1 (only the $(*)$ -algebra axioms appear) | v2.0: §0.1, Definition 0.1.2 + Remarks 0.1.3, 0.1.4

v1.0: "(no Lie–Jordan content; v1.0 §0.1 jumps from Def 0.1.1 directly to Def 0.1.2 Primitive Constraints)"

v2.0: " $AB = \frac{1}{2}[A, B] + (A \circ B) \dots [A, B] := AB - BA \dots A \circ B := \frac{1}{2}(AB + BA)$ "

Impact. The single largest conceptual addition to v2.0 §0. The decomposition equips A with two genuinely independent algebraic structures: a Lie algebra A_L (antisymmetric, ordering) and a Jordan algebra A_J (symmetric, co-occurrence). Remark 0.1.3 invokes Jordan–von Neumann–Wigner (1934) to assert their mathematical independence — the algebraic foundation of the two-primitive picture.

► NEW — Causality and locality named as structural co-primitives (§0.4)

Type: Added | v1.0: not present (v1.0 §0.4 is the Reconciliation Propagator R_t) | v2.0: §0.4, Definitions 0.4.1, 0.4.2, Theorem 0.4.4, Remarks 0.4.3, 0.4.5

v1.0: "The Reconciliation Propagator R_t is the dynamical engine of the framework."

v2.0: "the framework introduces two structural co-primitives that govern all subsequent emergence: causality and locality."

Impact. v2.0 §0.4 is an entirely new subsection that has no v1.0 §0 antecedent. Causality is grounded in the Lie algebra A_L (the "when"-capacity); locality is grounded in the Jordan algebra A_J (the "where"-capacity). Theorem 0.4.4 (Complementarity) replaces v1.0's Theorem 0.5.1 (Convergence) as the section's headline theorem.

► §0.2.2 redefined from "General Kinematic State" to "Admissibility"

Type: Modified + Renamed | v1.0: §0.2, Definition 0.2.2 — General Kinematic State (key correction vs Gen 2) | v2.0: §0.2, Definition 0.2.2 — Admissibility (Zero-Constraint Compatibility)

v1.0: “A kinematic state is ANY positive linear functional ω_{kin} on A with $\omega_{\text{kin}}(1) = 1$. The state is NOT required to satisfy $\omega_{\text{kin}}(M) = 0$.”

v2.0: “A state $\omega \in S(A)$ is admissible (or physical) if $\omega(\hat{C}_{\alpha} \dagger \hat{C}_{\alpha}) = 0$ for all $\alpha \in I$, equivalently, $\omega(M) = 0$.”

Impact. v1.0's Def 0.2.2 emphasized that the kinematic state is unconstrained (the methodological correction vs Gen 2). v2.0 moves this unconstrained-state emphasis into the surrounding prose and reclaims Def 0.2.2 to define admissibility — the target of the dynamics. This sharpens the kinematic/admissible distinction and pushes the dynamics that connects them (R_t , Convergence Theorem) into §1, where it can be defined without circularity.

► v1.0 §0.4 Reconciliation Propagator R_t removed (moved to v2.0 §1)

Type: Removed (moved to §1) | v1.0: §0.4 (entire subsection, Defs 0.4.1–0.4.4 including SOE/MOE/Dephasing mechanisms) | v2.0: moved to §1 (per v2.0 preamble: “Everything that depends on the fracture — R_t ... moves to Section 1”)

v1.0: “The Reconciliation Propagator R_t is the dynamical engine of the framework. It is not a single flow but the ordered composition of three scale-distinct mechanisms”

v2.0: “Everything that depends on the fracture — R_t , the Convergence Theorem, A_{red} , M_{red} , the Dirac bracket, burden as a physical quantity, and the Reconciliation Principle — moves to Section 1”

Impact. The single largest content migration. v1.0's §0.4 — roughly 1.5 pages of three-mechanism dynamics (SOE spectral-flux flow, MOE Bures-metric descent, dephasing) — leaves §0 entirely. v1.0's “ $\delta\hat{C}$ loop fix” badge, which patched a circularity between §0.4 and §1.1.1, is no longer needed because §0.4 is gone.

► v1.0 §0.8 Reconciliation Principle removed — moved to v2.0 §2 (not §1)

Type: Removed (moved to §2) | v1.0: §0.8 (Canonical Statement + Migration Notes; Principle 0.8.1, Theorem 0.8.2, Corollary 0.8.3) | v2.0: moved to §2 (per v2.0 §0.5 deferred-objects table: “Reconciliation Principle (variational) → §2”)

v1.0: “For any set S of mutually correlated events, the system evolves under R_t toward a state that minimizes the total relational inconsistency”

v2.0: “Reconciliation Principle (variational) → §2 — Requires burden, which requires the fracture”

Impact. v1.0 had already relocated RP from §0.4 (Gen 3) to §0.8 (its own canonical form) using a “forward-reference contract” that admitted the dependency on §1 (causal order $<$) and §2 (correlation kernel K_{ω}). v2.0 finishes the migration: RP leaves §0 entirely and goes to §2, where both Cubic ingredients are available. This removes the last forward reference from §0.

► NEW Remark 0.3.7 — Algebraic Availability vs. Physical Interpretation

Type: Added | v1.0: §0.3 contains no such meta-remark | v2.0: §0.3, Remark 0.3.7

v1.0: “Its expectation value $B_{\Delta}[\rho] = \text{Tr}(\rho F)$ is the obstruction burden (Definition 0.3.7).”

v2.0: “ F is algebraically available at Layer Q : the definition requires only objects defined in §0.1–0.3. However, F 's physical interpretation ... requires the Open Expansion machinery of Section 1.”

Impact. v2.0 introduces an explicit two-level distinction — algebraic availability at §0 vs. physical interpretation at §1 — that v1.0 did not make. This is the conceptual hinge that lets v2.0 §0 retain F while shedding every object that depends on F 's physical meaning (burden, R_t , A_{red}). The same distinction licenses the deferred status of Π_{net} and the reverse kernel inclusion.

► NEW — Acyclicity Test section and "What §0 Does NOT Contain" table

Type: Added | v1.0: §0.9 Architectural Summary contains no negative-content table and no acyclicity test | v2.0: §0.5 Architectural Summary, with the new "What Section 0 Does NOT Contain (Deferred to §1 or Later)" table and the "Acyclicity Test" subsection

v1.0: “The emergence ladder is acyclic: no layer depends on a layer below it, and no definition is used before its ingredients exist.”

v2.0: “Verdict: Section 0 is acyclic. The architecture holds at the foundational layer.”

Impact. v2.0 §0 closes with an explicit negative-space inventory (13 deferred objects, each with its target section and the reason for deferral) and a verdict-style acyclicity check. v1.0's §0.9 had no such inventory — it could not have listed A_{red} or M_{red} as "not contained" because v1.0 §0 did contain references to them (without ever defining them).

Section Impact Assessment

A reader of v2.0 §0 encounters a section that is structurally inverted relative to its v1.0 counterpart. Where v1.0 §0 was a dynamical engine — building R_t in §0.4, deriving Master-Zero in §0.5, splitting A_{phy} in §0.6–0.7, defining the Dirac bracket in §0.7b, and stating RP in §0.8 — v2.0 §0 is a naming ceremony: it names the algebra, the Lie–Jordan decomposition, admissibility, the GNS Hilbert space, F as an algebraic object, and the two co-primitives causality and locality, then stops. The headline theorem changes from Theorem 0.5.1 (Convergence to the Physical Sector, with proof sketch and $M_6(C)$ toy model) to Theorem 0.4.4 (Complementarity of the Governing Mechanisms, with no proof sketch). The reader leaves v2.0 §0 with the algebraic substrate in hand but no dynamics — dynamics is explicitly the property of §1.

The forward references that v1.0 §0 carried — and explicitly defended via a "forward-reference contract" box in the preamble — are eliminated. v1.0 §0.3.8 (Π_{net}) forward-referenced K_ω from §2; v1.0 §0.8 (RP) forward-referenced both $<$ from §1 and K_ω from §2; v1.0's A_{red} and M_{red} were referenced by §4 but never defined inside §0 (structural issue (ii) in the v2.0 preamble). v2.0 §0 contains none of these objects, and the Acyclicity Test at the end of §0.5 makes the absence explicit and checkable. The circularity that v1.0 patched with the $\delta\hat{C}$ replacement (the " $\Delta\hat{C}$ LOOP FIX" badge) is dissolved rather than fixed: §0.4 is gone, so there is no §0.4 → §1.1.1 dependency loop to fix.

Key Quotes from v2.0 §0

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

“This is a structural rewrite of Section 0 against the reordered emergence chain.”

“Everything that depends on the fracture — R_t , the Convergence Theorem, A_{red} , M_{red} , the Dirac bracket, burden as a physical quantity, and the Reconciliation Principle — moves to Section 1”

“The test for this section: does it contain anything that depends on fracture, sub-Hilbert spaces, burden, R_t , A_{red} , M_{red} , the Dirac bracket, or the Reconciliation Principle? If yes, the reordering has failed.”

“The central architectural commitment of this section is the two-primitive picture, grounded in the Lie-Jordan decomposition of the algebra.”

“Verdict: Section 0 is acyclic. The architecture holds at the foundational layer.”

4.2 v2.0 §1 — Open Expansion, Fracture, and the Reconciliation Propagator

Change type: REBORN | Changes identified: 18 | v2.0 size: 40,625 chars

Mapping

v2.0 §1 is a REBORN section assembled by requisitioning the entire Quadratic/Quartic content of v1.0 §0 and merging it with the Causal-layer content of v1.0 §1, then re-spinning the combined material around a single new architectural primitive — the fracture of H_{kin} . From v1.0 §0, v2.0 §1 absorbs: the Reconciliation Propagator R_t and its three-mechanism decomposition (v1 §0.4), the Convergence Theorem and Master-Zero equivalence plus $\Lambda_B = 0$ (v1 §0.5), the thin/full A_{phy} split with its fixed-point characterization (v1 §0.6 + §0.7 + Theorem 0.7.3), the algebraic and emergent Dirac bracket (v1 §0.7b), the obstruction burden B_Δ and burden linearity (v1 §0.3.7), and the physical interpretation of the fracture operator F (whose algebraic definition remains in v2 §0).

Structural Changes

From v1.0 §1, v2.0 §1 inherits the Open Extension Principle (v1 §1.4.3, restated as the Open Expansion Principle), the zero-preserving event definition (v1 §1.1.1), the primitive causal relation $<$ (v1 §1.1.3), and the emergent speed limit $c = \gamma \cdot \ell_0$. It does NOT inherit the two-scale depth formula $d(E) = d_{\text{SOE}} + N \cdot d_{\text{MOE}}$ (collapsed), the Two-Link Principle (v1 §1.5, moved to v2 §3), or the Emergent Direction / Lorentz compatibility material (v1 §1.7, moved to v2 §3).

Newly written content with no v1 antecedent dominates the first half of the section: the Open Expansion Principle framed as joint Lie-Jordan activation (Post 1.1.1), the Lie-Jordan mismatch (Def 1.1.3), the fracture of H_{kin} into sub-Hilbert spaces (Def 1.2.1), F 's physical interpretation as fracture measure (Thm 1.2.4), burden re-narrated as "cost of the fracture" (Def 1.3.1), and the Tier 1/Tier 2 separation made into a first-class definition (Def 1.4.1). The most consequential new construction is Definition 1.4.3 — M_{red} as the Tier 2 sub-collective of M , framed as "separation, not reduction" — which the v2.0 preamble explicitly identifies as resolving a v1.0 gap: " M_{red} was referenced but never defined."

The emergence chain is reordered to eliminate the residual circularity that v1.0 tried to patch with the $\delta\hat{C}$ replacement and the thin/full split. v1.0's chain still had two structural faults: RP at §0.8 was stated before burden's physical interpretation was available, and Π_{net} (§0.3.8) forward-referenced K_ω from §2. v2.0 §1 collapses this to a single-section chain: Open Expansion → finite rate → Lie-Jordan mismatch → fracture → F interpretation → burden → Tier 1/Tier 2 → A_{red} , M_{red} , Dirac → R_t → Convergence → A_{phy} split → events → causal order. RP and Π_{net} are deferred; burden is given its physical meaning before it is used in R_t .

Top Changes (with Before/After Quotes)

► Section title and identity reborn

Type: Renamed / Reborn | v1.0: §0 ("Algebraic Foundation, Reconciliation Propagator") + §1 ("Causal Foundation, Zero-Preserving Skeleton") | v2.0: §1 ("Open Expansion, Fracture, and the Reconciliation Propagator")

v1.0: "Section 1 / Causal Foundation / Zero-Preserving Skeleton"

v2.0: "Section 1 — Open Expansion, Fracture, and the Reconciliation Propagator"

Impact. v1.0 §1's identity was "causal foundation"; v2.0 §1's identity is "fracture-and-reconciliation." The causal-order material that was v1.0 §1's headline content is now its final subsection (§1.8), and the propagator material that was v1.0 §0's headline content becomes v2.0 §1's centerpiece (§1.5–1.6). The rebirth is structural, not cosmetic.

► SOE/MOE terminology renamed

Type: Renamed | v1.0: §0.4.1 ("SOE Spectral-Flux Flow"), §0.4.2 ("MOE Gradient Descent"); §1 preamble ("SOE/MOE two-scale") | v2.0: §1.5.2 ("Open Expansion Flow"), §1.5.3 ("Open Extension Flow")

v1.0: "the SOE (Single-Open-Extension) spectral-flux flow ... the MOE (Multiple-Open-Extension) gradient descent"

v2.0: "SOE is renamed Open Expansion — the local, isometric expansion of locality-built sectors. MOE is renamed Open Extension"

Impact. The most visible terminology change in the section. The v2.0 preamble justifies the rename: "Single Open Extension implied discrete, isolated extension events; Open Expansion captures the continuous, driven expansion of sectors toward reconciliation." The mathematics is unchanged; the physical interpretation is sharpened. The Definition 1.5.2/1.5.3 headers explicitly parenthesize "(was SOE)" and "(was MOE)" for traceability.

► NEW — Fracture of H_{kin} into sub-Hilbert spaces (Definition 1.2.1)

Type: Added | v1.0: (closest antecedent: v1 §0.8 Corollary 0.8.3 "fracture event," but there H_{kin} fractures as an output of RP, not as a precondition) | v2.0: §1.2.1 (Definition 1.2.1)

v1.0: "(v1.0 §0 has no fracture-of- H_{kin} concept independent of RP)"

v2.0: " $H_{kin} \rightarrow \oplus_k H_k$... Each H_k is a sector where the Jordan structure has locally converged toward Lie-Jordan compatibility."

Impact. This inverts the v1.0 logical order. In v1.0, the fracture was a downstream consequence of the Reconciliation Principle (Corollary 0.8.3). In v2.0, the fracture is upstream — it is what gives F its physical interpretation (§1.2.4), what creates the burden (§1.3), and what makes the Tier 1/Tier 2 split visible (§1.4). The RP itself is deferred to §2. This inversion is the single largest architectural change in v2.0 §1.

► NEW — M_{red} as Tier 2 sub-collective ("separation, not reduction")

Type: Added | v1.0: (no formal definition; v1 §0.7b uses A_{red} with $[\cdot, \cdot]_D$ but never defines M_{red} as an operator on A_{red}) | v2.0: §1.4.3 (Definition 1.4.3) + Remark 1.4.4

v1.0: "(v1.0 references M_{red} implicitly via A_{red} but never provides a definition)"

v2.0: " $M_{red} = \Sigma_{\{i \in I_2\}} w_i \cdot D_i^\dagger D_i \in A_{red}$... M_{red} is NOT a 'reduction' of M ; it is the SEPARATION of M into its Tier 1 and Tier 2 sub-collectives"

Impact. This is the most consequential new construction. The v2.0 preamble explicitly flags it: " M_{red} was referenced but never defined (it requires the fracture and the Tier 1/Tier 2 split)." The framing "separation, not reduction" is also new — it pre-empts a possible reader misreading of M_{red} as a "reduced" version of M . Remark 1.4.4 ties $\ker(M_{red})$ to the field admissibility condition in §2.

► Reconciliation Propagator R_t migrated from §0.4 → §1.5

Type: Relocated | v1.0: §0.4 (Definition 0.4.4 + Lemma 0.4.5) | v2.0: §1.5 (Definition 1.5.1)

v1.0: “ $R_t = \lim_{N \rightarrow \infty} (R_{T/N}^{\wedge \text{MOE}} \circ (R_{\varepsilon}^{\wedge \text{SOE}})^{\wedge N}), T = N \cdot \varepsilon$ ”

v2.0: “ $R_t = \lim_{N \rightarrow \infty} (R_{T/N}^{\wedge \{\text{OExt}\}} \circ (R_{\varepsilon}^{\wedge \{\text{OExp}\}})^{\wedge N}), T = N \cdot \varepsilon$ ”

Impact. The mathematical form is identical (only the SOE/MOE labels are renamed to OExp/OExt). The relocation places R_t after the burden, Tier 1/Tier 2 split, A_{red} , M_{red} , and Dirac bracket — i.e. after the fracture machinery that justifies the propagator's three-mechanism decomposition. In v1.0, R_t appeared at §0.4 before any of its conceptual ingredients existed.

► Reconciliation Principle REMOVED from §0 (deferred to v2 §2)

Type: Removed (from §1's scope; relocated to v2 §2) | v1.0: §0.8 (Reconciliation Principle, variational form) | v2.0: §2 (per §1 preamble; not present in v2 §1)

v1.0: “PRINCIPLE (Reconciliation, variational form). For any set S of mutually correlated events, the system evolves under R_t toward a state that minimizes the total relational inconsistency $I(S)$ ”

v2.0: “The Reconciliation Principle (variational) — §2 — Needs burden (available) + correlational structure (§3) for full target”

Impact. This is the deepest conceptual shift. v1.0 placed RP at §0.8, treating it as the capstone of the foundation. v2.0 moves RP out of §0 and out of §1 entirely, into §2, because RP's variational target $I(S)$ requires the correlation strength $s(a,b)$ — an algebraic precursor to K_{ω} that has no business in §1. The v2.0 §1 preamble explicitly states the v1.0 fault: “the Reconciliation Principle was stated before burden's physical interpretation was available.”

► Open Extension Principle → Open Expansion Principle (reframed as Lie-Jordan activation)

Type: Modified / Renamed | v1.0: §1.4.3 (Postulate, Open Extension Principle) | v2.0: §1.1.1 (Postulate 1.1.1, Open Expansion Principle)

v1.0: “A physical relational structure may continue only through admissible zero-preserving extensions: $\text{Ext}(_s) \neq \emptyset$ for all physical $_s$.”

v2.0: “local sectors (Jordan-built structures within H_{kin}) continuously expand their algebraic scope to reconcile with the single global causal layer (Lie structure).”

Impact. v1.0's principle was a combinatorial openness postulate on extension histories. v2.0's principle is a dynamical principle grounded in the Lie/Jordan decomposition: sectors must continuously expand to maintain joint Lie-Jordan compatibility. The activation of the two structural primitives (causality/Lie, locality/Jordan) becomes the foundational dynamic, replacing v1.0's “extendible zero-consistent partial order” framing.

► NEW Theorem 1.6.3 — Local Convergence, Global Desynchronization

Type: Added | v1.0: (no direct antecedent; v1 §1.2.3 has “Two-Scale Causal Structure” but is purely about causal order) | v2.0: §1.6.3 (Theorem 1.6.3)

v1.0: “(v1.0 has no formal theorem of local-conv/global-desync)”

v2.0: “ $\rho_{\infty} = \oplus_k p_k \cdot \rho_{\infty}^{\wedge \{k\}} + \rho_{\text{cross}} \dots$ The residual cross-sector burden ... appears as dark energy.”

Impact. This theorem gives the local-convergence / global-desynchronization structure its first formal statement. It is the algebraic seed of the dark-energy mechanism (Section 6) and of the deferred “now” (Section 4). v1.0 had the conceptual ingredients (two-scale structure, finite-rate propagation) but never assembled them into this theorem.

Section Impact Assessment

The acyclic emergence chain is achieved by a single device: every object that v1.0 had placed in §0 before its conceptual ingredients existed is moved into v2.0 §1 and placed after those ingredients. The ingredients in question are the fracture (§1.2), the burden (§1.3), and the Tier 1/Tier 2 split (§1.4). Once those are in place, A_{red} and M_{red} are definable (§1.4.2–1.4.3), the Dirac bracket is licensable (§1.4.5), R_t can be composed (§1.5), the Convergence Theorem closes the reverse kernel inclusion (§1.6), the thin/full A_{phy} split is available (§1.7), and zero-preserving events plus the operational causal order $<$ finally arrive (§1.8). The chain is short, monotone, and provably acyclic — the closing Acyclicity Test verifies each link's prerequisites by section number.

The conceptual shift from "RP at §0.8 before burden" to "RP after burden is defined" is the single most consequential reordering. In v1.0, the Reconciliation Principle was the capstone of the foundation (§0.8) — its variational target $I(S)$ consumed K_ω (forward-referenced from §2) and $<$ (forward-referenced from §1.1) while burden was still an algebraic quantity at §0.3.7 with no physical meaning. In v2.0, the RP is moved out of §0 and out of §1 into §2; burden receives its physical interpretation (§1.3, "cost of the fracture") before R_t is composed (§1.5) and before the Convergence Theorem (§1.6). This means RP can be stated in §2 against a fully grounded burden, rather than against an algebraic placeholder.

Key Quotes from v2.0 §1

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

" M_{red} was referenced but never defined (it requires the fracture and the Tier 1/Tier 2 split)"

" M_{red} is NOT a 'reduction' of M ; it is the SEPARATION of M into its Tier 1 and Tier 2 sub-collectives"

"the Reconciliation Principle was stated before burden's physical interpretation was available"

"Open Expansion activates the two primitives → finite rate creates a Lie-Jordan mismatch → the mismatch fractures H_{kin} into sub-Hilbert spaces"

"The residual cross-sector burden — the cost of maintaining this desynchronized composite — is the quantity that, at cosmological scale (Section 6), appears as dark energy."

4.3 v2.0 §2 — Burden, Matter, and the Reconciliation Principle

Change type: CONSOLIDATED | Changes identified: 20 | v2.0 size: 40,625 chars

Mapping

v2.0 §2 is the largest consolidation in the framework: it takes objects v1.0 had distributed across four sections at four emergence layers (§0.8 Quartic, §3 temporal, §4 matter, §7 probability) and rehomes them inside one section framed as the QM layer — the exact microscopic dynamics between the fracture (v2 §1.2) and correlational geometry (v2 §3). v1.0 had interleaved these objects with the GR-layer derivations: §3 supplied both burden and $\alpha(B)$, §4 supplied both fields and the emergent spatial arena ($X_{\omega, \text{red}}$, $D_{\omega, \text{red}}$). v2.0 pushes burden upstream into §1.3, the emergent metric downstream into §3, and keeps only microscopic-QM content (RP, dark energy, fields, particles, mass, records, Born, FIREWALL) in §2 — a self-contained QM chapter that is coarse-grained into GR in §3–§6.

Structural Changes

Each v1.0 source contributes a distinct sub-assembly. From §0.8 comes the variational RP (now §2.1, with $s(a,b)$ defined locally as an algebraic precursor rather than as a forward reference to §2.1's K_{ω}). From §4 come the field concept (Def 4.1 → Def 2.3.1), the covariant correlation Laplacian, the stable-mode and particle-like-excitation definitions, the Particle Identity theorem, and the mass-burden identity. From §7 come the record sector, the intra/cross-sector burden, stable record separation, redundant encoding, the sectorwise zero-decomposition, Z-envariance, the Born rule, and the FIREWALL. v1.0 §3 contributes only context — its burden $B(R)$ and $\alpha(B)$ are now in v2 §1.3 and v2 §4, and the §3.2.5 forward reference is closed by Cor 2.5.2.

The newly written material falls into three categories. First, three structural fixes advertised in the Preamble: (i) RP moved out of §0.8 because it had been stated before burden's interpretation existed; (ii) the mass-burden identity $m \equiv B_0$ was algebraically inconsistent with $B_0 = m^2$ in Eq 4.2.2 and is now stated as $m^2 \equiv B_0$; (iii) Theorem 7.4 (sectorwise zero-decomposition) was content-free and is now restated with M_{red} (properly defined at v2 §1.4.3). Second, two new definitions that close v1 forward references: the algebraic correlation strength $s(a,b)$ (Def 2.1.2) and the local burden flux $J_{\{ij\}}(B)$ (Def 2.3.3). Third, an entirely new dark-energy mechanism (§2.2) that derives $\rho_{\text{DE}} \sim (H/\Gamma)^2$ with $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$, predicts the 120-order suppression as $(\ell_{\text{P}}/R_{\text{H}})^2$, and predicts early-dark-energy suppression $\rho_{\text{DE}} \sim (1+z)^{-3}$.

Top Changes (with Before/After Quotes)

► Reconciliation Principle relocated from §0.8 to §2.1

Type: Relocated + modified | v1.0: §0.8.1 (Principle 0.8, variational form) — Layer Q' , after §0.5–§0.7b | v2.0: §2.1 (Principle 2.1.1) — after burden (§1.3) and events (§1.8)

v1.0: “In the canonical manuscript, RP must be stated as a Layer Q' (Quartic) principle, immediately after Section 2 has supplied its Cubic ingredients (causal order $<$, correlation kernel K_{ω}).”

v2.0: “The Reconciliation Principle was stated at §0.8, before burden's physical interpretation was available. It now sits at §2.1, after burden (§1.3) and events (§1.8) are defined.”

Impact. Eliminates the v1.0 forward-reference contract from §0.8 to §§1–2 (the contract that made §0.8 a "principle awaiting its target"). In v2 RP is stated with burden, events, $\prec$, and $s(a,b)$ all in hand — the single biggest architectural correction.

► NEW dark-energy mechanism (the major addition)

Type: Added (no v1.0 antecedent for the quantitative mechanism) | v1.0: §8.3.3 had only the qualitative claim "dark energy = dynamic pressure of open extension"; §3.1.4 had listed "relational burden" as a quarantined candidate for vacuum energy | v2.0: §2.2 (Defs 2.2.1, Prop 2.2.2, Conj 2.2.3, Eqs 2.2.4–2.2.5)

v1.0: "(v1.0 §8.3.3 qualitative only — "dynamic residual burden pressure of open extension")"

v2.0: " $\rho_{DE}(t) \sim (H(t) / \Gamma_{eff}(t))^2 \cdot \rho_{source} \dots \Gamma_{eff}(t) \sim \eta \cdot B_{\Delta}(t) \dots$ The quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch ($B_{\Delta} = \|\text{mismatch}\|^2$)."

Impact. Introduces restorative power $\mathcal{P}_{restore} = \eta \cdot \|\nabla_{\{\text{Bures}\}} B_{\Delta}\|^2$, the burden-dependent relaxation rate $\Gamma_{eff} \sim \eta \cdot B_{\Delta}$, and the dark-energy density scaling $\rho_{DE} \sim (H/\Gamma)^2 \cdot \rho_{source}$. The 120-order suppression is $(H_0 \cdot t_P)^2 \sim 10^{-122}$ because the exponent 2 is structural (from $B_{\Delta} = \|\text{mismatch}\|^2$), not tuned. The mechanism is established in §2; the cosmological application is deferred to §6.

► Mass-burden identity corrected ($m \equiv B_0 \rightarrow m^2 \equiv B_0$)

Type: Modified (algebraic fix) | v1.0: §4.2.8 (P3 + Eq 4.2.2 + Remark 4.2) — inconsistent triple | v2.0: §2.5.1 (Thm 2.5.1)

v1.0: " $m \equiv B_0$ (P3 headline, but Eq 4.2.2 says $B_0 := B(\text{supp}(\varphi_0)) = \lambda_0 = m_{eff}^2$)"

v2.0: "The squared mass of a particle-like excitation is the ground-state maintenance burden: $m^2 \equiv B_0 \dots$ equivalently, $m = \sqrt{B_0}$ in natural units, or $m = (\hbar_{eff} / c_{eff}) \cdot \sqrt{B_0}$ in dimensionful units."

Impact. Resolves the v1.0 algebraic inconsistency. The three-step derivation ($m_{eff}^2 = \lambda_0$; $\lambda_0 = B_0$; hence $m^2 = B_0$) is consistent throughout, with a consistency-check table confirming all four statements line up. The dimensionful coefficient ($\hbar_{eff}$) is honestly flagged as open pending the §5 continuum-limit proof.

► Sectorwise zero-decomposition fixed with M_{red} (Thm 7.4 → Thm 2.7.1)

Type: Modified (content fix) | v1.0: §7.4.2 (Thm 7.4) — used $\mathcal{M}_B = \sum p_i \hat{C}_i^\dagger \hat{C}_i$ with arbitrary p_i | v2.0: §2.7.1 (Thm 2.7.1)

v1.0: "(v1.0 Thm 7.4 was content-free — $\omega(\mathcal{M}_B) = 0$ with arbitrary p_i)"

v2.0: "Theorem 7.4 (sectorwise zero-decomposition) was content-free — introducing $\mathcal{M}_B = \sum p_i \hat{C}_i^\dagger \hat{C}_i$ and noting $\omega(\mathcal{M}_B) = 0$ did not constrain the p_i . The rewrite fixes this by stating the decomposition in terms of M_{red} (now properly defined at §1.4.3)."

Impact. The most subtle technical fix. v1.0's Thm 7.4 was tautological — the weights p_i were free parameters and $\omega(\mathcal{M}_B) = 0$ was satisfied trivially. v2's restatement uses $M_{red} = \oplus_k M_{red}^{\wedge\{k\}} + M_{red}^{\wedge\{\text{cross}\}}$; positivity of M_{red} plus decoherence suppression forces $\omega_k(M_{red}^{\wedge\{k\}}) = 0$ for each k with $p_k > 0$. The weights are then constrained by normalization and sectoral structure — not arbitrary.

► Born rule strengthened by M_{red} objectivity

Type: Modified | v1.0: §7.5.5 (Thm 7.5) — five assumptions including "branch measures are objective zero-closure invariants (Theorem 7.4)" | v2.0: §2.7.4 (Thm 2.7.4)

v1.0: "(v1.0 Thm 7.5 objectivity premise was vacuous because Thm 7.4 was content-free)"

v2.0: “The unique normalized branch measure consistent with: (1) Phase invariance (from Z-invariance) ... (3) Objectivity (from the sectorwise zero-decomposition, Theorem 2.7.1 — the weights are determined by M_{red} 's sectoral structure, not arbitrary) ... is the Born rule: $p_i = |c_i|^2$.”

Impact. The v1.0 objectivity assumption (Thm 7.4) was content-free, so the v1.0 Born rule derivation inherited a vacuous premise. With Thm 2.7.1 now genuinely constraining p_k via M_{red} 's sectoral decomposition, the Born rule's "objectivity" premise has real content. The five-step argument is restated with additivity explicitly sourced to "the linearity of M_{red} 's sectoral decomposition."

► Particle definition tightened from bounded-burden to low-burden

Type: Modified | v1.0: §4.2.4 (Def 4.5) — four conditions: zero-preservation, localisation, stability, identifiability | v2.0: §2.4.2 (Def 2.4.2)

v1.0: “The mode is stable if, as the relational structure undergoes open extension, the constraint burden required to maintain the Tier 2 zero-preservation of φ_n remains bounded: $B(\text{supp}(\varphi_n)) \leq B_{max} < \infty$.”

v2.0: “The v1.0 definition required only $B \leq B_{max}$ (bounded burden). The corrected definition requires additionally that the mode be a local minimum of the burden functional (low burden). This distinguishes particles (cheap stable modes) from other stable configurations (e.g., neutron stars, which are stable but high-burden).”

Impact. Tightens the particle concept — particles are now local minima of the burden functional, not merely bounded-burden modes. This physically distinguishes elementary particles from high-burden stable configurations (neutron stars, black holes).

► NEW Cross-Sector Gravity Quarantine

Type: Added | v1.0: (none — v1.0 §7.6 had nine guardrails but no explicit quarantine of cross-sector gravity) | v2.0: §2.8.2 (Guardrail 2.8.2)

v1.0: “(v1.0 has no explicit cross-sector gravity quarantine)”

v2.0: “Cross-sector gravity — the hypothesis that distinct sectors gravitationally interact through their branch weights — is *QUARANTINED*. It would require either: (a) A probabilistic average over sectors sourcing curvature (*PROHIBITED* by the FIREWALL), or (b) An algebraic cross-sector burden coupling not currently present in the framework (would require a new primitive).”

Impact. Explicitly quarantines speculative gravity theories involving cross-sector gravitational interaction via branch weights. Notes that the relational burden channel $T^{(rel)} = [\hat{C}_\alpha, \Pi_{net}]$ (the derived dark-matter mechanism, deferred to §4) does NOT cross the FIREWALL because it is algebraic.

► Particle Identity theorem demoted to definitional equivalence

Type: Modified | v1.0: §4.2.7 (Thm 4.1) | v2.0: §2.4.4 (Thm 2.4.4)

v1.0: “(v1.0 Thm 4.1 stated as a derived theorem)”

v2.0: “The v1.0 Theorem 4.1 was tautological (its hypotheses were its conclusion). The rewrite makes the equivalence explicit and honest — it is a definitional equivalence, not a derived theorem.”

Impact. Removes a false-positive theorem. v2 admits the four defining properties ARE the theorem's hypotheses and explicitly labels the result as a definitional equivalence.

Section Impact Assessment

The consolidation changes the framework's logical flow in three ways. First, it removes v1.0's interleaving of microscopic-QM content with coarse-grained-GR content: where v1.0 had $\text{burden} + \alpha(B)$ in §3 (with temporal geometry), $\text{fields} + \text{particles} + \text{mass}$ in §4 (alongside $X_{\omega, \text{red}}$), and $\text{records} + \text{Born} + \text{FIREWALL}$ in §7 (after cosmology §8), v2.0 collects all QM-layer objects in §2 and pushes all GR-layer objects (metric, lapse, arrow of time, burden tensor) into §3–§4. Second, it eliminates v1.0's most-cited forward references — the §0.8 → §2.1 K_{ω} forward reference (closed by Def 2.1.2), the §4.1.3 → §3.1.4 burden-flux phantom reference (closed by Def 2.3.3), and the §3.2.5 → §4.2.8 mass-burden forward reference (closed by Cor 2.5.2). Third, it sharpens the section's identity: v2 §2 is unambiguously "the QM layer — the exact microscopic dynamics that precedes the coarse-grained GR perspective."

The new dark-energy mechanism (§2.2) is the most consequential cosmological addition. The mechanism is structurally forced by $\Lambda_B = 0$ (Cor 1.6.4) plus the existence of the Open Extension flow: if there is no vacuum energy, observed cosmic acceleration must be the rate of an ongoing process — and the only ongoing process available is the relaxation of accumulated open-expansion burden. The 120-order suppression falls out as $(H_0 \cdot t_P)^2 \sim 10^{-122}$ because the exponent 2 is structural (from $B_{\Delta} = \|\text{mismatch}\|^2$), not tuned. The Γ -was-larger-in-the-past prediction gives $\rho_{DE} \sim (1+z)^{-3}$ and $\Omega_{DE} \sim (1+z)^{-6}$, predicting $\Omega_{DE} \sim 10^{-18}$ at recombination — resolving the early-dark-energy tension without fine-tuning.

Key Quotes from v2.0 §2

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

"The reordering consolidates them into a single section because they all belong to the same emergence layer — the exact microscopic dynamics where quantum mechanics lives."

"The Reconciliation Principle was stated at §0.8, before burden's physical interpretation was available. It now sits at §2.1, after burden (§1.3) and events (§1.8) are defined."

"The mass-burden identity $m \equiv B_0$ was algebraically inconsistent with the section's own equations ($B_0 = m^2$ in Eq 4.2.2, but $m \equiv B_0$ in the headline). The rewrite states the identity consistently as $m^2 \equiv B_0$."

"Theorem 7.4 (sectorwise zero-decomposition) was content-free — introducing $\mathcal{M}_B = \sum p_i \hat{C}_i \dagger \hat{C}_i$ and noting $\omega(\mathcal{M}_B) = 0$ did not constrain the p_i ."

" $\rho_{DE}(t) \sim (H(t) / \Gamma_{\text{eff}}(t))^2 \cdot \rho_{\text{source}} \dots$ The quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch ($B_{\Delta} = \|\text{mismatch}\|^2$)."

4.4 v2.0 §3 — Emergent Geometry

Change type: RENUMBERED | Changes identified: 14 | v2.0 size: 30,527 chars

Mapping

v2.0 §3 (Emergent Geometry) is a RENUMBERED version of v1.0 §2 (Emergent Space) — "largely intact in content" per the Preamble, with four targeted corrections and the relocation of Π_{net} from v1.0 §0.3.8 into v2.0 §3.9. The mathematical spine is preserved: the pairwise kernel K_{ω} , the phase-preserving kernel K_{ω} , the cubic kernel $K_{\omega}(A,B,C)$, the logarithmic distance $d_{\omega} = -\ell_0 \log K_{\omega}$, Cubic Factorization + Cubic Dominance, the quotient metric (X_{ω}, d_{ω}) , displacement profiles $\Delta_{\{x \rightarrow y\}}$, the dimensional closure argument, and the Type-Sign Coupling lemma. The $L \rightarrow Q \rightarrow C \rightarrow Q'$ emergence ladder that dominated v1.0 §2.0.1 is removed; in its place §3.0 gives a four-paragraph "Purpose: The Operational 'Where'" tying §3 to the now-closed QM layer of §2.

Structural Changes

The four advertised corrections are surgical. (1) $\Xi_C(D)$ is redefined as rank deficiency of the $D \times D$ cubic Gram matrix plus $(D - 3)_+$ (replacing v1.0's tautological $|D - 3|$), and Thm 3.6.4 is demoted from "Theorem" to "[Conditional Proposition — v1.0 'Theorem' demoted]". (2) The volume element V_{ω} is recomputed from K_{ω} instead of K_{ω} , with Remark 3.1.5 calling out "the v1.0 error ... is corrected here". (3) The Type-Sign Coupling remains necessary with sufficiency deferred to §4, but the T-4 dual-use issue is resolved — T-4 is now reserved for the Born rule (closed in §2.7). (4) A new Conjecture 3.8.2 "Smooth Field Convergence" grounds the continuum limit in stable field-mode density (§2.4) rather than an abstract averaging operation.

The Π_{net} relocation is the third structural move. v1.0 §0.3.8 placed Π_{net} in §0, forward-referencing K_{ω} (then in §2.1). v2.0 §3.9 introduces $\Pi_{\text{net}} = \sum \{E_i\} P_{\{E_i\}} \cdot K_{\omega} \cdot P_{\{E_i\}}$ "now possible because K_{ω} and the sectoral decomposition are both available," with Remark 3.9.2 closing the forward reference and rerouting the §4 relational dark-matter channel $T^{\text{rel}} = [\hat{C}_{\alpha}, \Pi_{\text{net}}]$ to reference §3.9.

Top Changes (with Before/After Quotes)

► D=3 closure proof circularity fix and theorem demotion

Type: Modified + Renamed (status demoted) | v1.0: §2.7.3 (Thm 2.7.2) | v2.0: §3.6.3 (Def 3.6.3) + §3.6.4 (Thm 3.6.4) + Remark 3.6.5

v1.0: " $\Xi_{\text{under}}(D) = 3 - D > 0 \dots \Xi_{\text{over}}(D) = D - 3 > 0 \dots$ Combining all three cases: $\Xi_C(D) = 0 \Leftrightarrow D = 3$."

v2.0: " $\Xi_C(D) := \text{rank deficiency of the } D \times D \text{ cubic Gram matrix} + (D - 3)_+ \dots$ This is NOT defined as $|D - 3|$ (the v1.0 tautological definition)."

Impact. $\Xi_C(D)$ is now non-tautological, but Thm 3.6.4 is explicitly relabeled "[Conditional Proposition — v1.0 'Theorem' demoted]." The framework's most distinctive prediction is honestly downgraded, not salvaged.

► Phase-preserving kernel K_{ω} used in the relational volume element

Type: Modified (correction of internal contradiction) | v1.0: §2.7.2 (Def 2.7.1) | v2.0: §3.6.2 (Def 3.6.2)

v1.0: " $V_{\omega}(A, B, C) := \det [K_{\omega}^{\wedge(3)}]$ where $K_{\omega}^{\wedge(3)}_{ij} = K_{\omega}(X_i, X_j)$."

v2.0: “ $V_\omega(A, B, C) = |\det [K_\omega(X_i, X_j)]| \dots$ The v1.0 version used K_ω (magnitude-only), which cannot correctly detect linear dependence.”

Impact. Closes v1.0's internal contradiction — §2.1.2 warned of "a 96% false-positive rate" for magnitude-only Gram determinants, yet §2.7.2 used exactly that kernel. The corrected V_ω uses the complex Gram determinant, so the linear-independence premise of D=3 closure is at least coherent.

► Type-Sign Coupling retained as necessary; T-4 dual-use resolved

Type: Modified (status tracking, not mathematics) | v1.0: §2.8.2 (Thm 2.8.2) | v2.0: §3.7.2 (Thm 3.7.2)

v1.0: “the signature $(-, +, +, +)$ is structurally forced as a necessary condition ... sufficiency is a Theorem Target (T-4 in §9 audit), deferred to Section 5.”

v2.0: “This resolves the v1.0 T-4 dual-use issue. The Lorentzian-signature sufficiency is no longer called 'T-4' (which is reserved for the Born rule, closed in §2.7).”

Impact. Mathematical content unchanged (necessary condition established, sufficiency deferred), but T-4 is freed for reuse by the Born rule and the deferred target is rerouted from "§5 Gravity" (v1.0) to "§4" (v2.0).

► Smooth Field Convergence introduced as complementary coarse-graining mechanism

Type: Added | v1.0: (none — coarse graining was purely triangle-inequality-based via §2.5 + §2.8.3) | v2.0: §3.8 (Conjecture 3.8.2)

v1.0: “(v1.0 has no smooth-field-convergence concept)”

v2.0: “As the density of stable field modes on (X_ω, d_ω) increases, the discrete correlation structure converges to a smooth field configuration. The smooth field IS the smooth manifold.”

Impact. The coarse-graining bridge now has two complementary mechanisms — triangle inequality (metricity) and smooth field convergence (smoothness). Tagged [Structural conjecture].

► Π_{net} relocated from §0.3.8 to §3.9 (forward reference closed)

Type: Added (relocation from v1.0 §0.3.8) | v1.0: §0.3.8 (forward-referenced K_ω from §2.1 — an explicit circularity) | v2.0: §3.9.1 (Def 3.9.1) + Remark 3.9.2

v1.0: “ $\Pi_{\text{net}} = \sum [E_i] P_{[E_i]} K_\omega P_{[E_i]} \dots$ where $P_{[E_i]}$ are projectors onto open extensions E_i and K_ω is the correlation kernel (defined in Section 2.1).”

v2.0: “ $\Pi_{\text{net}} = \sum [E_i] P_{[E_i]} \cdot K_\omega \cdot P_{[E_i]} \dots$ Π_{net} is derivable from F and K_ω — zero new primitives.”

Impact. Closes a v1.0 forward reference; Π_{net} is now defined after K_ω (§3.1) and the sectoral projectors (§1.2, §2.1.5) are both available. The §4 dark-matter channel $T^{\wedge}(\text{rel}) = [\hat{C}_\alpha, \Pi_{\text{net}}]$ now references §3.9, making the dependency graph acyclic.

► Orientation Invariant O_ω dropped entirely

Type: Removed | v1.0: §2.7.4 (Def 2.7.3, “QUARANTINED”) | v2.0: absent

v1.0: “ $O_\omega(A, B, C) := \text{sign}(\text{Im}[K_\omega(A, B) \cdot K_\omega(B, C) \cdot K_\omega(C, A)]) \dots$ strictly quarantined as a structural conjecture in Section 9.”

v2.0: “(dropped — no replacement, no Preamble comment)”

Impact. v1.0's quarantined three-body chirality invariant is dropped from §3 with no replacement. The phase-preserving kernel K_ω that fed O_ω is retained (and elevated), but the orientation use-case is gone.

► Open Target 1 / §1.3.5 / §1.7.3 certification callouts removed

Type: Removed | v1.0: §2.3.5 ("Open Target 1 — CLOSED" box + "§1.3.5 now CERTIFIED" box), §2.4 ("§1.7.3 now CERTIFIED" box), §2.9 (P8 box) | v2.0: absent (Remark 3.3.4 replaces the ceremony)

v1.0: "Open Target 1 — CLOSED. Theorem 2.3.3 closes Open Target 1 (metricity) of the Construction Spec without abandoning the logarithmic distance."

v2.0: "The v1.0 'Open Target 1 — CLOSED' label overstated the status. The rewrite acknowledges: the triangle inequality is derived conditional on these conditions holding."

Impact. v1.0's certification ceremony (Open Target 1, P1–P8 patches, forward-reference callouts) is stripped. Remark 3.3.4 concedes v1.0's label "overstated the status." The triangle inequality is honestly tagged [Established (conditional)].

Section Impact Assessment

The $D=3$ circularity fix is the most consequential change because $D=3$ is the framework's "most distinctive prediction," per Remark 3.6.5 itself. v1.0's $\Xi_C(D) = |D - 3|$ made the theorem a definitional tautology — the conclusion was baked into the premise. v2.0's corrected $\Xi_C(D) = (\text{rank deficiency of the } D \times D \text{ cubic Gram matrix}) + (D - 3)_+$ is genuinely non-tautological, but Thm 3.6.4 is explicitly tagged "[Conditional Proposition — v1.0 'Theorem' demoted]" and Remark 3.6.5 concedes the $D=3$ claim "still relies on the structural assumptions (i) and (ii), which are not derived from the algebra." The fix is therefore honest rather than salvific: the framework's signature dimensional claim is no longer defended as a theorem, and the rewrite acknowledges that a skeptic can still ask "why can't there be a 4th inference channel?" without an algebraic answer.

The phase-preserving kernel fix makes the corrected closure argument at least coherent. The Gram determinant is the standard linear-algebraic tool that detects linear dependence via determinant rank, and a magnitude-only kernel destroys the complex phase information this rank-detection relies on. v1.0 §2.1.2 explicitly warned of "a 96% false-positive rate" for magnitude-only Gram determinants, then proceeded to use exactly that magnitude-only kernel in §2.7.2's volume element — an internal contradiction. Replacing K_ω with $\tilde{K}_\omega$ in Def 3.6.2 ensures the linear-independence premise of the $D=3$ argument is at least mathematically well-posed; combined with the demotion of the theorem itself, the framework now claims less but claims it coherently.

Key Quotes from v2.0 §3

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

"The $D=3$ closure proof was circular — the closure defect $\Xi_C(D)$ was defined as $|D-3|$, making ' $\Xi_C(D) = 0 \iff D = 3$ ' a tautology."

"The relational volume element used the magnitude-only kernel, contradicting §2.1's own warning about Gram determinants. The rewrite uses the phase-preserving kernel $\tilde{K}_\omega$ for the Gram matrix."

" $\mathcal{IT}_{\text{net}}$ is derivable from F and $\tilde{K}_\omega$ — zero new primitives. It projects onto the subspace of observables correlated across multiple open-extension sectors."

“The $D=3$ closure is the framework's most distinctive prediction, but it is not a theorem in the rigorous sense. ... The rewrite acknowledges this.”

4.5 v2.0 §4 — Time, Gravity, and the Three-Tier Burden

Change type: MERGED | Changes identified: 17 | v2.0 size: 32,263 chars

Mapping

v2.0 §4 (Time, Gravity, and the Three-Tier Burden) is a MERGER of v1.0 §3 (Emergent Time) and v1.0 §5 (Gravity) — they "belong to the same emergence layer — the GR perspective (coarse-grained macroscopic dynamics)" per the Preamble. In v1.0, time (§3) and gravity (§5) were separated by the matter section (§4 Fields/Particles), which forced §3.2.5 to forward-reference §5 for the mass–burden identity and forced §5 to import burden, $\alpha(B)$, and the ADM lapse from §3 across a section boundary. v2.0 closes both seams by reordering: matter (now §2) sits upstream, the smooth manifold (§3.8) is step 17, and §4 is steps 18–20 — duration, gravity, three-tier burden — all in one block.

Structural Changes

The Three-Tier Burden (§4.4) is the conceptual reframing that gives the section its title. v1.0 §5.1.2 called this the "Three-Channel Decomposition" with explicit activation weights $a_{\text{mode}}, a_{\text{int}}, a_{\text{rel}} \in [0,1]$ (Def 5.2) — three independently selectable components. v2.0 §4.4 drops the activation weights entirely and reframes the channels as "three facets of the single reconciliation process (burden)" (Remark 4.4.2). The renaming from "channel" to "tier" reinforces this: the tiers are not separable substances but simultaneous aspects. Remark 4.4.2 also introduces a NEW "Coincidence Problem" argument — because mode/int/rel all derive from one burden, their densities "are correlated by construction," structurally addressing why dark matter, dark energy, and baryonic matter share order-of-magnitude densities today.

What stays from v1 §3 in v2 §4: the burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$, the burden-weighted proper time $\tau[\gamma]$, the burden-clock potential Φ_C , the weak-burden expansion $\Phi_C \approx -c^2_{\text{eff}} \cdot \lambda \cdot B$, and the Effective Burden Metric Ansatz. What stays from v1 §5: the burden tensor $\Theta^\wedge(B)_{\mu\nu}$ as variational derivative of B_Δ , the three-tier decomposition with relational burden as derived dark-matter mechanism, the ADM dictionary, metric boundedness $\det(h) \geq \ell_0^{\wedge\{2d\}}$, $\Lambda_B = 0$, $\kappa_B = C/(\Pi_{\text{max}} \cdot \ell_0^2)$, the Newtonian limit, and the relational-burden halo.

Top Changes (with Before/After Quotes)

► Arrow-of-Time generator swap corrected

Type: Modified (corrected proof) | v1.0: §3.6.1 Theorem 3.6.1 | v2.0: §4.1.4 Theorem 4.1.4

v1.0: "The SOE component (§0.4.1) is generated by $-\gamma[F, [F, \rho]]$, a Lindblad-type double commutator"

v2.0: "The v1.0 proof swapped the Open Expansion and Open Extension generators...and used a Klein-inequality argument that yielded 0 identically"

Impact. v1's generator assignment contradicted §0.4 (where SOE/Open-Expansion is Hamiltonian and MOE/Open-Extension is dissipative). v2 restores the correct mapping — Open Extension = Lindblad with $L = F$, Open Expansion = Hamiltonian $i[\sqrt{(F+\delta F)}, \rho]$ — and replaces the Klein inequality (which yielded 0 because ρ and $\log \rho$ commute) with Spohn's inequality for Lindblad entropy production.

► Newtonian-limit sign error fixed

Type: Modified (sign correction) | v1.0: §5.5.1 Theorem 5.7 | v2.0: §4.6.1 Theorem 4.6.1

v1.0: “The Einstein tensor component $G_{00} \approx -2\nabla^2\Phi/c^2$ in the weak-field limit”

v2.0: “The v1.0 had $G_{00} \approx -2\nabla^2\Phi/c^2$, which is wrong for the $(-, +, +, +)$ signature (Carroll §4; Weinberg). The correct identity is $G_{00} \approx +2\nabla^2\Phi/c^2$.”

Impact. v1's sign was internally inconsistent — the proof derived $\nabla^2\Phi = +4\pi G \cdot B(x)$ (correct conclusion) from a G_{00} identity with the wrong sign, masking the error. v2 cites Carroll §4 and Weinberg and carries the corrected sign through. The $C = 8\pi$ calibration is now performed here (in the Newtonian limit), not in the saturation argument.

► Einstein-closure circularity broken (conservation theorem)

Type: Modified (non-circular proof) | v1.0: §5.2.1 Theorem 5.2 | v2.0: §4.3.4 Theorem 4.3.4

v1.0: “Assume the effective metric $g_{B\dots}$ satisfies a candidate geometric equation of the form: $G_{\mu\nu}[g_B] + \Lambda_B \cdot g_{\mu\nu}(B) = \kappa_B \cdot \Theta_{\mu\nu}(B)$ ”

v2.0: “The conservation follows from the diffeomorphism invariance of $B_\Delta[\rho]$ (Noether-style argument), NOT from assuming the Einstein field equation.”

Impact. v1 Thm 5.2 assumed the candidate field equation as a premise — but that same equation was the conclusion of Thm 5.5 four subsections later, making conservation circular. v2 replaces the Bianchi-identity derivation with a Noether-style argument from diffeomorphism invariance of the scalar functional $B_\Delta[\rho]$. The circularity is genuinely broken; conservation now precedes and is independent of the closure.

► Einstein-closure status reverted Strengthened → Conditional

Type: Downgraded | v1.0: §5.4.1 Theorem 5.5 | v2.0: §4.5.4 Theorem 4.5.4

v1.0: “Einstein-Like Closure (P5 — Strengthened)”

v2.0: “[Conditional]. The closure is conditional on the continuum limit (T-2). The v1.0 'Strengthened' status is reverted — the key premise (smooth 4D Lorentz-compatible manifold) is unproven.”

Impact. v1 advertised the closure as Strengthened via “two independent derivation paths.” v2 reverts to Conditional, acknowledging the key premise (smooth 4D Lorentz-compatible continuum limit) is open target T-2. This is the honest-downgrade counterpart to the genuine circularity fix in the conservation theorem: the circularity is broken, but the closure itself is still not unconditionally proven.

► κ_B derivation made honest (C calibrated, not derived)

Type: Downgraded (circularity acknowledged) | v1.0: §5.4.3 Theorem 5.6, P7 | v2.0: §4.5.6 Theorem 4.5.6

v1.0: “ $\kappa_B = C / (\Pi_{\max} \cdot \ell_0^2)$...where C is a dimensionless geometric factor of order unity (analogous to 8π in standard GR)”

v2.0: “The v1.0 claimed $C = 8\pi$ 'consistent with the dimensional analysis.' This is circular: the saturation argument gives $C \sim 1$ (order unity), not $8\pi \approx 25.13$. The value $C = 8\pi$ is CALIBRATED by the Newtonian limit”

Impact. v1 P7 advertised κ_B as “DERIVED from saturation limit” while identifying C with 8π “analogous to standard GR” — but the saturation argument only fixes C to order unity ($8\pi \approx 25.13$ is not $O(1)$). v2 splits the claim: the dimensional form $\kappa_B \propto 1/(\Pi_{\max} \cdot \ell_0^2)$ is derived; the dimensionless factor $C = 8\pi$ is calibrated by the Newtonian limit.

► NEW "Now" definition + Global Desynchronization

Type: Added | v1.0: absent (deferred from §0 in v1) | v2.0: §4.2 Definitions 4.2.1, Theorems 4.2.2–4.2.3, Remark 4.2.4

v1.0: “(no v1 antecedent — the "now" was deferred from §0 and never defined)”

v2.0: “The 'now' at an event $e \in E_{phy}$ is the joint definition of where and when at the same instance: $Now(e) = (x(e), \tau(e))$ ”

Impact. v2 §4.2 introduces a wholly new architectural object: the "now" as the joint product of operational "where" (§3 space) and operational "when" (§4.1 duration), with the Principle of the Now (Thm 4.2.2) tying it to Lie-Jordan compatibility. Global Desynchronization (Thm 4.2.3) yields a NEW dark-energy interpretation: "Dark energy is the cost of the global desynchronization of local nows."

► Type-Sign Sufficiency — T-4 dual-use resolved

Type: Modified (open target closed) | v1.0: §2.8.2 Type-Sign Coupling (necessary); T-4 dual-use tracked as open | v2.0: §4.5.3 Theorem 4.5.3

v1.0: “(v1 tracked Lorentzian-signature sufficiency as Theorem Target T-4, dual-used with Born rule)”

v2.0: “This resolves the v1.0 T-4 dual-use issue. The Lorentzian-signature sufficiency is no longer tracked as 'T-4' (reserved for the Born rule, closed in §2.7).”

Impact. v1 had a single label T-4 doing double duty (Type-Sign sufficiency for GR + Born rule objectivity), creating a dependency tangle. v2 splits them: Born rule closes T-4 in §2.7; Type-Sign Sufficiency is established here by uniqueness — no other signature is consistent with (i) total causal order, (ii) spatial pseudometric d_ω , (iii) burden-clock suppression $\alpha(B)$ simultaneously.

► Three-Tier Burden — activation weights dropped + Coincidence Problem

Type: Modified (reframing) + Added | v1.0: §5.1.2 Definition 5.2 | v2.0: §4.4 Definition 4.4.1 + Remark 4.4.2

v1.0: “ $a_{mode}, a_{int}, a_{rel} \in [0,1]$ are activation weights representing the coupling of each channel to the effective geometry (component activation is model-dependent)”

v2.0: “The three channels are not separate substances; they are three facets of the single reconciliation process (burden)...different aspects of the same burden, so their densities are correlated by construction.”

Impact. v1's Def 5.2 introduced tunable activation weights suggesting the channels were independently selectable. v2 §4.4 drops the weights entirely — the decomposition is now exact, not weighted — and reframes "channels" as "tiers." NEW Remark 4.4.2 leverages this to address the dark-matter/dark-energy/baryonic-matter coincidence problem: densities are correlated by construction.

Section Impact Assessment

The Arrow-of-Time generator swap is the most consequential proof correction in §4. v1 §3.6.1 assigned the Lindblad-type double commutator $-\gamma[F, [F, \rho]]$ to SOE and the dissipative Bures gradient $-\eta[M, \rho]$ to MOE, contradicting §0.4 where SOE (Open Expansion) is the Hamiltonian isometric flow and MOE (Open Extension) is the dissipative descent. The v1 proof then invoked the Klein inequality $dS/dt = \eta \cdot \text{Tr}([M, \rho] \cdot \log \rho) \geq 0$, which yields 0 identically because ρ and $\log \rho$ commute — so the v1 "Arrow" was structurally vacuous. v2's corrected proof (Thm 4.1.4) uses the actual generators (Open Extension = Lindblad with $L = F$, Open Expansion = Hamiltonian $i[\sqrt{(F+\delta F)}, \rho]$) and invokes Spohn's inequality for Lindblad entropy production, giving non-trivial $dS/dt \geq 0$ with equality iff ρ commutes with $L = F$. This is a genuine fix, not a relabeling.

The Einstein-closure circularity is genuinely broken but the closure itself is honestly downgraded, not unconditionally proven. Change 5 fixes the circularity in the conservation theorem by deriving $\nabla^\mu \Theta^\mu(B)_{\mu\nu} = 0$ from diffeomorphism invariance of B_Δ (Noether-style), independent of the field equation. But Change 6 reverts the closure itself from Strengthened to Conditional, acknowledging the smooth 4D Lorentz-compatible continuum-limit premise (T-2) is open. Change 7 additionally corrects the "two independent derivation paths" claim — both paths cash out through Lovelock, so they are not mutually reinforcing. Net effect: the circularity between Thms 5.2 and 5.5 is dissolved, but the closure remains Conditional (T-2).

Key Quotes from v2.0 §4

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

“consolidates v1.0 §3 (Emergent Time) and §5 (Gravity) into a single section because they belong to the same emergence layer — the GR perspective”

“The v1.0 proof swapped the Open Expansion and Open Extension generators...and used a Klein-inequality argument that yielded 0 identically”

“The v1.0 had $G_{00} \approx -2\nabla^2 \Phi/c^2$, which is wrong for the $(-, +, +, +)$ signature (Carroll §4; Weinberg).”

“The conservation follows from the diffeomorphism invariance of $B_\Delta[\rho]$ (Noether-style argument), NOT from assuming the Einstein field equation.”

“The v1.0 'Strengthened' status is reverted — the key premise (smooth 4D Lorentz-compatible manifold) is unproven.”

“The v1.0 claimed $C = 8\pi$ 'consistent with the dimensional analysis.' This is circular: the saturation argument gives $C \sim 1$ (order unity), not $8\pi \approx 25.13$.”

4.6 v2.0 §5 — Black Holes

Change type: RENUMBERED | Changes identified: 14 | v2.0 size: 23,577 chars

Mapping

v2.0 §5 (Black Holes) is a RENUMBERED version of v1.0 §6 (Black Holes — Unreconciled Relational Sectors) — "largely intact in content" per the Preamble, with six targeted corrections and a layer-badge fix. The substantive content survives essentially unchanged: the same seven conceptual moves (gravity basin vs. black-hole domain, ℓ_0 -floor singularity avoidance, dimensional suppression to 2D, dual horizon ratios, boundary recovery, entropy-area scaling, lowest-burden emission) appear in the same order, and the nine interpretive guardrails carry over with only minor rewording.

Structural Changes

What changes is the framing apparatus. v1.0's patch system (P1–P7), per-subsection epistemic tags, per-subsection source citations, page footers, four summary tables, and the verbose dependency-contract preamble are all removed. In their place, v2.0 §5 carries a single compact preamble enumerating the six corrections, an Acyclicity Test section, and a two-table summary in §5.8 (a 16-row "Contains" table and a 5-row "Does NOT Contain (Deferred)" table). The theorem count shifts from 7 theorems + 7 definitions to 6 theorems + 5 definitions + 1 principle + 1 conjecture.

The six explicit corrections concentrate where v1.0 overclaimed. Two fix outright errors: the missing " $\Rightarrow$ " symbol in Eq. (6.4.3) that inverted Theorem 6.3's claim, and the "Layer C (Cubic)" badge that violated §0's layer spine. Two downgrade overclaimed results: $\Pi_{\max}$'s "RESOLVED" → "partially resolved" (with a new dimensional Definition 5.1.3); the Bekenstein-Hawking "T-6 partially closed" → "T-6 area-scaling only; coefficient open," with explicit acknowledgment that the v1.0 argument was circular. One correction fixes seven stale cross-references. The sixth demotes two tautological theorems (6.6 → Principle 5.6.1; 6.7 → Conjecture 5.6.2), each noting the v1.0 proof "restated the conclusion." v2.0 §5 is thus a de-escalation of v1.0 §6: same architectural claims, lower epistemic temperature.

Top Changes (with Before/After Quotes)

► Missing " $\Rightarrow$ " in Eq. (6.4.3) — Fixed

Type: Fixed | v1.0: §6.4.2, Eq. (6.4.3) | v2.0: §5.4, Theorem 5.4.2, Eq. (5.4.2)

v1.0: " $\varepsilon R \rightarrow 0 < R = \emptyset$. (no negation arrow — reads as " $\varepsilon_R \rightarrow 0$ implies $<_R = \emptyset$ ")"

v2.0: " $\varepsilon_R \rightarrow 0 \Rightarrow <_R = \emptyset$ (5.4.2)"

Impact. The most damaging single error — the displayed equation literally asserted the opposite of Theorem 6.3's claim and Guardrail 7. The fix is flagged inline ("The v1.0 Eq. (6.4.3) was missing the ' $\Rightarrow$ ' symbol, literally reading ... the opposite of the theorem's claim").

► Layer-Badge Spine Violation — Fixed

Type: Fixed / Renamed | v1.0: §6.0–§6.7 subheaders ("LAYER C (MOE scale)") | v2.0: §5.2–§5.6 subheaders ("LAYER Q' · QUARTIC (COARSE-GRAINED MOE HYDRODYNAMICS)")

v1.0: "Section 6 occupies Layer C exclusively, like Section 5"

v2.0: "Section 5 is badged Layer Q' (Quartic — coarse-grained MOE hydrodynamics), not 'Layer C (Cubic).' Per §0, the Cubic layer is supplied by Sections 1–3; Sections 4–5 are Quartic."

Impact. v1.0 badged §6 as "Layer C" but per its own §0 layer spine, the Cubic layer is supplied by §§1–3 (microscopic structure), while §§4–5 are Quartic (coarse-grained hydrodynamics). The fix corrects a self-incoherence in the L→Q→C→Q' emergence ladder. v1.0's "P1 — Black holes belong EXCLUSIVELY to Layer C" patch is dropped entirely, subsumed into the Layer Q' badge.

► $\Pi_{\max}$ Formal Definition Added; "RESOLVED" Downgraded

Type: Added / Downgraded | v1.0: §6.1 P2 patch ("§5.3.3 forward reference RESOLVED") | v2.0: §5.1, Definition 5.1.3 + Remark 5.1.4

v1.0: "The §5.3.3 forward reference is RESOLVED."

v2.0: " $\Pi_{\max} := C_{\Pi} / \ell_0^2$... The forward reference from §4.5.6 ... is now PARTIALLY RESOLVED ... The dimensionless ratio C/C_{Π} remains to be fixed by the spectral analysis (T-2)."

Impact. v1.0 declared the forward reference "RESOLVED" without ever defining $\Pi_{\max}$. v2.0 adds Definition 5.1.3 ($\Pi_{\max} = C_{\Pi} / \ell_0^2$) derived dimensionally from the burden-stress dimensions $[\text{length}]^{-2}$ and the ℓ_0 -floor as the only available microscopic length scale, with C_{Π} flagged as requiring T-2. Status: "PARTIALLY RESOLVED — form closed, coefficient open."

► Bekenstein-Hawking Coefficient Circularity Acknowledged

Type: Downgraded | v1.0: §6.5.1, Theorem 6.5 ("T-6 partially closed") | v2.0: §5.5, Theorem 5.5.3

v1.0: "This is the structural derivation of the Bekenstein-Hawking entropy formula $S = AH/(4\ell P_2)$... The Bekenstein-Hawking coefficient $1/(4\ell P_2)$ is recovered ..."

v2.0: "The Bekenstein-Hawking coefficient $1/(4\ell P^2)$ is NOT derived. The argument above assumes the exponential growth of microstates (which IS the conclusion, restated). The identification $\ell_P \equiv \ell_0$ fixes the units of S but not the dimensionless prefactor $1/4$."

Impact. Most consequential honesty correction. v1.0 assumed exponential microstate growth (the conclusion) and "derived" area scaling. v2.0 separates the established result (area scaling, which follows from dimensional suppression) from the open result (the dimensionless $1/4$ prefactor, which requires a microscopic count of zero-preserving boundary modes). Status moves from "T-6 partially closed" to "T-6 area-scaling only; coefficient open."

► Tautological Theorems Demoted to Principle and Conjecture

Type: Downgraded / Renamed | v1.0: §6.5.2 Theorem 6.6 (Lowest-Burden Emission); §6.5.3 Theorem 6.7 (Hawking-Like Emission) | v2.0: §5.6 Principle 5.6.1; §5.6 Conjecture 5.6.2

v1.0: "Theorem 6.6 (Lowest-Burden Emission). Let R be a strongly burden-loaded region undergoing relaxation. The emitted sector is not the original infallen matter class, but the lowest-burden admissible output..."

v2.0: "PRINCIPLE 5.6.1 — Lowest-Burden Emission [Structural principle; NOT a theorem] ... The v1.0 'Theorem 6.6' was tautological — its proof restated the conclusion ..."

Impact. v2.0 explicitly identifies both v1.0 theorems as tautological. Theorem 6.6 → Principle 5.6.1 (a variational assumption, not a derivation). Theorem 6.7 → Conjecture 5.6.2 (thermal appearance plausible but unproven; β not derived). For the conjecture, v2.0 lists two concrete open tasks: prove detailed balance, and derive β from saturation parameters. $T_H = 1/\beta$ is now explicitly "not derived."

► Stale Cross-References Corrected (Seven Anchors)

Type: Fixed | v1.0: §6.0–§6.5 inline references | v2.0: §5.3, §5.4, §5.5 inline references + §5.8 Acyclicity Test

v1.0: “references to “Definition 2.10,” “Theorem 2.8,” “Definition 2.9,” “Theorem 4.2.2,” “Theorem 0.6.3,” “Principle 1.1,” “Definition 3.7””

v2.0: “Definition 2.10 → Definition 3.6.2; Theorem 2.8 → Theorem 3.6.4; Definition 2.9 → Definitions 3.5.3–3.5.4; Theorem 4.2.2 → Theorem 2.5.1; Theorem 0.6.3 → Theorem 1.7.4; Principle 1.1 → Theorem 0.4.4; Definition 3.7 → Theorem 4.1.1.”

Impact. v1.0 referenced seven anchors that no longer exist in the renumbered canonical form. v2.0 corrects all seven and notes one subtlety: “Theorem 4.2.2 did not exist in v1.0 (§4.2.2 was a subsection, not a theorem); Theorem 0.6.3 was renumbered to 0.7.3, which in the rewrite is Theorem 1.7.4.”

Section Impact Assessment

The layer-badge fix is more than cosmetic. By badging §5 as Layer Q' (Quartic) rather than Layer C (Cubic), v2.0 restores the coherence of the L→Q→C→Q' emergence ladder defined in §0. v1.0 had a self-contradiction: §0 placed the Cubic layer at §§1–3 (microscopic structure), but §6 declared itself “Layer C exclusive,” placing black holes in the same layer as the kinematic substrate. This was a logical problem: black holes are coarse-grained hydrodynamic phenomena, not microscopic structure. The v2.0 fix correctly situates §5 in the Quartic layer alongside §4 (Time/Gravity), consistent with the actual content.

The Bekenstein-Hawking circularity admission is the most consequential epistemic change. v1.0 §6.5 Theorem 6.5 claimed to “structurally derive” $S = A_H/(4\ell_P^2)$ and to “recover” the 1/4 coefficient by identifying ℓ_P with ℓ_0 . As v2.0 explicitly acknowledges, this was circular: the argument assumed exponential microstate growth (the conclusion) and then derived the area scaling. The v2.0 Theorem 5.5.3 honestly separates the established result (area scaling $S \propto \text{Area}(\partial R)$, which follows from dimensional suppression) from the open result (the dimensionless 1/4 prefactor, requiring a microscopic count of zero-preserving boundary modes). Status moves from “T-6 partially closed” to “T-6 area-scaling only; coefficient open.” The framework still derives the holographic boundary (Theorem 5.3.2) and area scaling (Theorem 5.5.3) non-circularly; what it does not derive is the BH coefficient.

Key Quotes from v2.0 §5

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

“This section is the v1.0 Section 6 largely intact in content, with six corrections.”

“The v1.0 Eq. (6.4.3) was missing the ‘ ϵ ’ symbol, literally reading ‘ $\epsilon_R \rightarrow 0 \prec_R = \emptyset$ ’ — the opposite of the theorem's claim.”

“ $\Pi_{\max} := C_{\Pi} / \ell_0^2$... The v1.0 ‘RESOLVED’ claim overstated the status. The rewrite acknowledges: $\Pi_{\max}$ is defined dimensionally ..., but the dimensionless coefficient C_{Π} requires T-2.”

“The Bekenstein-Hawking coefficient $1/(4\ell_P^2)$ is NOT derived. The argument above assumes the exponential growth of microstates (which IS the conclusion, restated).”

“Section 5 is badged Layer Q’ (Quartic — coarse-grained MOE hydrodynamics), not ‘Layer C (Cubic).’ Per §0, the Cubic layer is supplied by Sections 1–3; Sections 4–5 are Quartic.”

“T-6 area-scaling only; coefficient open.”

4.7 v2.0 §6 — Cosmology

Change type: RENUMBERED | Changes identified: 16 | v2.0 size: 25,063 chars

Mapping

v2.0 §6 (Cosmology) is a RENUMBERED version of v1.0 §8 (Cosmology — The Open Universe) — "largely intact in architecture" per the Preamble, with seven targeted corrections and the dark energy mechanism properly linked to §2.2. v2.0 §6 preserves the seven-subsection skeleton of v1.0 §8 almost exactly: §6.0 ↔ §8.0, §6.1 ↔ §8.1, §6.2 ↔ §8.2, §6.3 ↔ §8.3, §6.4 ↔ §8.5, §6.5 ↔ §8.4, §6.6 ↔ §8.6, §6.7 ↔ §8.7. The conceptual chain is identical — Open Expansion frontier → cubic volume growth → Friedmann-like equation → three-part dark sector → early universe (ω_{kin} + inflation + sector formation) → $\Lambda_B = 0$ → status tables and guardrails.

Structural Changes

The seven corrections are applied surgically, not by rewriting the section. (1) The Friedmann equation is honestly relabeled a structural Conjecture (6.2.2) with the derivation path stated in four explicit steps (FLRW ADM ansatz → burden tensor → S_{eff} variation → $8\pi G/3$ calibration), and crucially acknowledges the $8\pi G/3$ coefficient is imported from standard GR. (2) Dark energy is no longer "SOE frontier pressure" hand-waved into existence; it is Conjecture 6.3.2 with explicit scaling $\rho_{\text{DE}} \sim (H/\Gamma_{\text{eff}})^2 \cdot \rho_{\text{source}}$, $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$, citing Property 2.2.2 from §2.2. (3) The Γ evolution argument becomes Theorem 6.3.4 with quantitative $(1+z)^{-3}$ density scaling and $(1+z)^{-6}$ fraction scaling at recombination. (4) Cross-reference misattribution is fixed — "Theorem 5.4" (used for both Einstein closure and singularity avoidance in v1.0) becomes Theorem 4.5.4 and Theorem 5.2.1. (5) The "Theorem 8.5.1" label for $\Lambda_B = 0$ is downgraded to Corollary 6.4.1. (6) The self-contradictory Q-9 status "CLOSED (conditional on T-2)" becomes "CONDITIONAL (on T-2)". (7) The layer badge is corrected to Layer Q' (Quartic — cosmological application of coarse-grained MOE hydrodynamics).

Top Changes (with Before/After Quotes)

► Friedmann equation — asserted conjecture → conjecture with derivation path

Type: Modified / honestly downgraded | v1.0: §8.2, Conjecture 8.2.2 | v2.0: §6.2, Conjecture 6.2.2

v1.0: "MOE gradient descent on the cosmological metric yields: $(\dot{a}/a)^2 = (8\pi G/3) \cdot \rho_B - k/a^2$... This is the Euler-Lagrange equation of MOE descent on the FLRW metric."

v2.0: "The v1.0 asserted this equation without derivation. The rewrite honestly states the derivation path but acknowledges the computation has not been performed. The $8\pi G/3$ coefficient is not derived from RCF primitives; it is imported from standard GR."

Impact. The most substantive correction. v1.0 presented the Friedmann form as a structural consequence with no computation; v2.0 enumerates the four derivation steps, admits none have been performed, and flags the non-local relational channel averaging as unresolved. The conjecture status is honestly retained rather than dressed up.

► Dark energy mechanism — "SOE frontier pressure" → $(H/T)^2$ scaling from §2.2

Type: Modified / properly linked | v1.0: §8.3.3, Conjecture 8.3.3 | v2.0: §6.3, Conjecture 6.3.2

v1.0: “ $\rho_{\Delta\text{eff}}(t) = \langle \rho_{\text{Bresidual}}(t) \rangle \rightarrow 0$ as $t \rightarrow \infty$... dynamic residual burden pressure of open extension (Conjecture 8.3.3).”

v2.0: “ $\rho_{\text{DE}}(t) \sim (H(t) / \Gamma_{\text{eff}}(t))^2 \cdot \rho_{\text{source}} \dots \Gamma_{\text{eff}}(t) \sim \eta \cdot B_{\Delta}(t)$ is the burden-dependent relaxation rate (Property 2.2.2) ... quadratic scaling $(H/\Gamma)^2$ comes from burden being quadratic in mismatch.”

Impact. v1.0 had no quantitative form — only a vague time-averaged residual. v2.0 imports the exact $(H/\Gamma)^2$ scaling from §2.2, grounds the quadratic exponent in $B_{\Delta} = \|\text{mismatch}\|^2$, and turns the hand-waved "frontier pressure" into a testable functional form.

► Γ evolution formalized — intuition → Theorem 6.3.4 with $(1+z)^{-6}$ suppression

Type: Added (formalized from qualitative intuition) | v1.0: §8.3.3 (implicit only) | v2.0: §6.3, Theorem 6.3.4

v1.0: “(v1.0 had no $\Gamma(z)$ argument)”

v2.0: “ $\rho_{\text{DE}} \sim (H / \Gamma_{\text{eff}})^2 \sim ((1+z)^{3/2} / (1+z)^3)^2 = (1+z)^{-3}$... The dark energy FRACTION $\Omega_{\text{DE}} \sim (1+z)^{-6}$, giving $\Omega_{\text{DE}} \sim 10^{-18}$ at recombination — eighteen orders of magnitude below the CMB bound ($\Omega_{\text{EDE}} < 0.004$).”

Impact. v1.0 had no $\Gamma(z)$ argument. v2.0 derives $(1+z)^{-3}$ for ρ_{DE} and $(1+z)^{-6}$ for Ω_{DE} using $H \sim (1+z)^{3/2}$ and $B_{\Delta} \sim \rho_{\text{matter}} \sim (1+z)^3$. This makes early dark energy suppression a theorem-target and explicitly resolves the CMB Ω_{EDE} bound.

► $\Lambda_B = 0$ — Theorem 8.5.1 → Corollary 6.4.1

Type: Downgraded / honestly labeled | v1.0: §8.5, Theorem 8.5.1 | v2.0: §6.4, Corollary 6.4.1

v1.0: “Theorem 8.5.1 ($\Lambda_B = 0$ is Exact). Conditional on T-2 (stable-mode assumption). Restates Theorem 5.5 in cosmological context.”

v2.0: “The v1.0 labeled this ‘Theorem 8.5.1’ despite being explicitly conditional on T-2. The rewrite is a COROLLARY (of Theorem 4.5.4 / Corollary 1.6.4), honestly conditional. A theorem conditional on an unproved conjecture is not a theorem; it is a conditional proposition.”

Impact. The most architecturally significant relabeling. v1.0's "Theorem" status was inconsistent with its own T-2 dependency. v2.0 codifies the rule that results conditional on open theorem-targets are corollaries, preserving "Theorem" for fully proven results.

► Q-9 status — "CLOSED (conditional on T-2)" → "CONDITIONAL (on T-2)"

Type: Fixed (logical contradiction resolved) | v1.0: §8.6.1, Table 8.6.1 ("Cosmological constant Λ " row); Q-9 box in §8.5 | v2.0: §6.4, Remark 6.4.2

v1.0: “Cosmological constant Λ ... CLOSED (conditional on T-2) — framework predicts $\Lambda = 0$ ”

v2.0: “The v1.0 ‘CLOSED (conditional on T-2)’ was self-contradictory — ‘CLOSED’ and ‘conditional on an open target’ cannot both hold. The rewrite uses ‘CONDITIONAL (on T-2)’ consistently, reserving ‘CLOSED’ for results with no open blocking dependencies.”

Impact. Propagates through the framework's status vocabulary. v2.0 reserves "CLOSED" for results with no open blocking dependencies, treating "conditional on an open target" as a distinct epistemic category.

► Cross-reference misattribution — "Theorem 5.4" → 4.5.4 / 5.2.1

Type: Fixed | v1.0: §8.2, §8.4.1 (and §8.6 guardrail 7) | v2.0: §6.2, §6.5, §6.7 acyclicity test

v1.0: “not a singularity (singularity avoidance is structural in RCF via the ℓ_0 -floor, Theorem 5.4 / Q-14) and the Einstein equation $G_{\mu\nu} = \kappa_B \Theta_{\mu\nu}(B)$ emerged as the Euler-Lagrange equation of MOE descent on the space of emergent metrics in §5.3 (Theorem 5.4)”

v2.0: “All stale cross-references corrected: ‘Theorem 5.4’ (for Einstein closure) → ‘Theorem 4.5.4’; ‘Theorem 5.4’ (for singularity avoidance) → ‘Theorem 5.2.1’; ‘Theorem 5.3.1 / 5.5’ → ‘Theorem 4.5.4 and Corollary 4.5.5’.”

Impact. v1.0 used “Theorem 5.4” for two distinct results plus the “Theorem 5.3.1 / 5.5” hybrid citation. v2.0 disambiguates each usage — a citation hygiene fix that materially improves traceability.

► Equation of state $w(z)$ — new Theorem 6.3.5 (falsifiable predictions)

Type: Added | v1.0: absent | v2.0: §6.3, Theorem 6.3.5

v1.0: “(v1.0 left $w \approx -1$ as a pending open question)”

v2.0: “ $w(t) = -1 - 2\dot{H} / (3H^2)$... Predictions: $w > -1$ always (no phantom crossing); w evolves from 0 (early) to -1 (late); $\rho_{DE} \rightarrow 0$ as $t \rightarrow \infty$ (decays, not constant) ... The prediction $w > -1$ always is distinct from both Λ CDM ($w = -1$ exactly) and quintessence.”

Impact. v1.0 left $w \approx -1$ as a pending open question. v2.0 derives an evolving $w(t)$ from the $(H/\Gamma)^2$ scaling and produces three testable predictions against DESI/Euclid, converting the dark energy story from descriptive to falsifiable.

Section Impact Assessment

The Friedmann equation is honestly downgraded, not derived. Conjecture 6.2.2 states the four-step derivation path explicitly (FLRW-restricted ADM ansatz → homogeneous/isotropic burden tensor components → S_{eff} variation → $\kappa_B = 8\pi G/c^4$ calibration by Newtonian limit) but admits “the computation has not been performed.” Two specific gaps are flagged: (a) the $8\pi G/3$ coefficient is imported from standard GR, not derived from RCF primitives; (b) the non-local relational channel Θ^{rel} (built from Π_{net} per §3.9) requires FLRW averaging that v1.0 never addressed and v2.0 leaves as part of the open derivation. The architectural cost is real: until the ADM-on-FLRW computation is performed, the entire Friedmann-level cosmology rests on a structural conjecture, and every downstream claim (Q-13 Λ CDM recovery, the $w(z)$ predictions) inherits that conditional status.

The dark energy mechanism linkage to §2.2 is the most consequential theoretical upgrade. v1.0’s “SOE frontier pressure” was a qualitative hand-wave with $\rho_{\Lambda\text{eff}}(t) = \langle \rho_{B\text{residual}}(t) \rangle \rightarrow 0$ and no scaling. v2.0’s Conjecture 6.3.2 imports $\rho_{DE} \sim (H/\Gamma_{\text{eff}})^2 \cdot \rho_{\text{source}}$ with $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}(t)$, then derives four downstream results: the 120-order resolution (Theorem 6.3.3, $\rho_{DE}/\rho_{\text{Planck}} = (H_0 \cdot t_P)^2 \approx 10^{-122}$), the early-dark-energy suppression (Theorem 6.3.4, $\rho_{DE} \sim (1+z)^{-3}$, $\Omega_{DE} \sim (1+z)^{-6}$), the equation of state (Theorem 6.3.5, $w > -1$ evolving from 0 to -1), and the asymptotic decay (Corollary 6.3.6, $\rho_{DE} \rightarrow 0$). The $(1+z)^{-6}$ suppression at recombination resolves a real observational tension: standard early-dark-energy models constrained by CMB data face the $\Omega_{\text{EDE}} < 0.004$ bound, and v2.0 predicts $\Omega_{DE} \sim 10^{-18}$ at $z \approx 1100$ — eighteen orders of magnitude below that bound.

Key Quotes from v2.0 §6

The following verbatim quotes from v2.0 best capture the section’s self-described changes from v1.0:

“The v1.0 asserted this equation without derivation. The rewrite honestly states the derivation path but acknowledges the computation has not been performed.”

“The $8\pi G/3$ coefficient is not derived from RCF primitives; it is imported from standard GR (with $\kappa_B = 8\pi G/c^4$ calibrated by the Newtonian limit).”

“ $\rho_{DE}(t) \sim (H(t) / \Gamma_{eff}(t))^2 \cdot \rho_{source} \dots \Gamma_{eff}(t) \sim \eta \cdot B_{\Delta}(t)$ is the burden-dependent relaxation rate (Property 2.2.2)”

“ $\Omega_{DE} \sim (1+z)^{-6}$, giving $\Omega_{DE} \sim 10^{-18}$ at recombination — eighteen orders of magnitude below the CMB bound ($\Omega_{EDE} < 0.004$).”

“A theorem conditional on an unproved conjecture is not a theorem; it is a conditional proposition.”

“The v1.0 'CLOSED (conditional on T-2)' was self-contradictory — 'CLOSED' and 'conditional on an open target' cannot both hold.”

4.8 v2.0 §7 — Audit, Formal Action, and Closure

Change type: RENUMBERED | Changes identified: 16 | v2.0 size: 38,155 chars

Mapping

v2.0 §7 (Audit, Formal Action, and Closure) is a RENUMBERED version of v1.0 §9 (Audit — Manuscript Closure) — "largely intact in architecture" per the Preamble, with six targeted corrections and a theorem-status table rebuilt wholesale. The skeleton is recognizably preserved: Purpose (§7.0) → Consolidated theorem table (§7.1) → Seven theorem targets T-1..T-7 (§7.2) → 17-point quarantine list (§7.3) → Formal action S_{eff} (§7.4) → Dependency graph + layer reconciliation (§7.5) → Guardrails + anti-patterns (§7.6) → Manuscript closure (§7.7) → Architectural summary (§7.8). However, the section title shifts from "Audit — Manuscript Closure" to "Audit, Formal Action, and Closure," elevating S_{eff} to title-level prominence.

Structural Changes

The six explicit corrections are: (1) the Theorem 9.1 sign/factor-of-2 error in the Einstein Closure derivation is fixed by adopting the standard GR convention $\Theta^{\wedge}(B)_{\mu\nu} = (2/\sqrt{g}) \delta/\delta g^{\mu\nu} \text{Tr}(\rho \cdot F_E)$; (2) the theorem-status table is rebuilt wholesale because v1.0 §9.1 had 6 of 9 Section 5 rows pointing to wrong section/theorem numbers; (3) the quarantine arithmetic is fixed (v1.0 claimed "13 PARTIAL" but listed 11 items, summing to $19 \neq 17$; v2.0 uses "11 PARTIAL" consistently); (4) the T-4 dual-use bug is resolved by reserving T-4 for the Born rule only and re-homing the Lorentzian-signature sufficiency as Theorem 4.5.3; (5) the M_{red} gap is closed by replacing nonexistent v1.0 references ("Thm 4.2.2", "Thm 0.6.3") with Theorem 1.7.4 (thin=full) and Theorem 2.5.1 (mass-burden identity), with M_{red} now defined at §1.4.3; (6) all stale cross-references are corrected throughout.

The 56% size reduction (~87K → ~38K chars) comes from surgical cuts, not content loss. Removed: the v1.0 §9.5.1 Conceptual Integrity Firewall table, the §9.5.2 Mechanism Assignment Summary (the entire SOE/MOE/dephasing column retires), the §9.5.4 Layer Assignment Audit, the §9.6.4 seven-pillar manuscript-closure narrative, the §9.7 conceptual-chain ASCII diagram with P1–P7 boxes, and the Section 0–8 cumulative merged-PDF list. Added in v2.0: a dedicated Preamble listing the six corrections, an explicit Status Taxonomy Definition (7.1.1), Remark 7.3.3 calling out the Q-9 status correction, Remark 7.4.4 honestly flagging λ_R as a new free parameter, a formally stated Theorem 7.5.1 (Acyclic Dependency Graph) where v1.0 only asserted acyclicity, a Forward-Reference Resolution Table 7.5.2, and Remark 7.5.3 retiring the A/B/C three-layer protocol in favor of $L \rightarrow Q \rightarrow C \rightarrow Q'$ only.

Top Changes (with Before/After Quotes)

► Theorem 9.1 sign/factor-of-2 fix (Einstein Closure) — Correction #1

Type: Fixed | v1.0: §9.4.2 Theorem 9.1 (Eq. 9.2) | v2.0: §7.4 Theorem 7.4.2 (Eq. 7.4.2)

v1.0: " $G_{\mu\nu} = \kappa_B \cdot TB_{\mu\nu}$ where $TB_{\mu\nu} = \delta/\delta g_{\mu\nu} \text{Tr}(\rho \cdot FE[g])$ is the burden tensor"

v2.0: " $G_{\mu\nu} = \kappa_B \cdot \Theta^{\wedge}(B)_{\mu\nu} \dots \Theta^{\wedge}(B)_{\mu\nu} := (2/\sqrt{g}) \cdot \delta/\delta g^{\mu\nu} \text{Tr}(\rho \cdot F_E)$ "

Impact. The v1.0 definition was off by a factor of 2 (and the implicit sign of the matter Lagrangian). v2.0 adopts the standard GR stress-energy convention $T_{\mu\nu} = -(2/\sqrt{g}) \delta S_{\text{matter}}/\delta g^{\mu\nu}$, so the variation of $(1/2\kappa_B)R$ and $-\text{Tr}(\rho \cdot F_E)$ closes to $G_{\mu\nu} = \kappa_B \Theta^{\wedge}(B)_{\mu\nu}$ correctly. This is the framework's central

GR-recovery theorem and was silently wrong in v1.0.

► Theorem-status table rebuilt against corrected numbering — Correction #2

Type: Rebuilt | v1.0: §9.1 Table 9.1.1 (~60 rows; 5 columns: Sec, Result, Status, Mechanism [SOE/MOE/dephasing], Depends On)
| v2.0: §7.1 Table 7.1.2 (~85 rows; 5 columns: Section, Result, Status, Mechanism, Depends On)

v1.0: "13 PARTIAL, 3 OPEN. Ports the formal action Seff from Ddd.pdf §3"

v2.0: "rebuilt against the corrected canonical numbering of Sections 0–6"

Impact. v1.0 had 6 of 9 Section 5 rows pointing to wrong section/theorem numbers (e.g., Thm 5.4 misattributed, Thm 5.5 misnumbered). v2.0 discards the SOE/MOE/dephasing mechanism tags in favor of descriptive mechanism phrases. Each row now points to a real §0.1–§6.5 location.

► Quarantine arithmetic fixed — Correction #3

Type: Fixed | v1.0: §9.3 Table 9.3.1 caption | v2.0: §7.3 Remark 7.3.2

v1.0: "1 CLOSED (Q-9), 2 STRUCTURALLY ADDRESSED (Q-7, Q-14), 13 PARTIAL ..., 3 OPEN"

v2.0: "1+2+13+3 = 19 ≠ 17, and the listed PARTIAL set had 11 items, not 13"

Impact. v1.0's arithmetic was internally inconsistent (19≠17). v2.0 uses corrected counts (0 CLOSED, 2 STRUCT, ~9–11 PARTIAL, ~6 OPEN) and admits the PARTIAL/OPEN boundary is fuzzy for Q-1 and Q-8, asking for a consistent convention. The 17-point list itself is preserved verbatim.

► T-4 dual-use bug resolved — Correction #4

Type: Fixed | v1.0: §9.2 Table 9.2.1 (T-4 = Born rule) AND §§1.7.4, 2.8.2, 5.4.1 (T-4 = Lorentzian sufficiency) | v2.0: §7.2 Remark 7.2.3 + Theorem 4.5.3

v1.0: "T-4 Born rule CLOSED in §7 (STRENGTHENED)"

v2.0: "T-4 is reserved exclusively for the Born rule. ... Lorentzian-signature sufficiency is now established as Theorem 4.5.3"

Impact. v1.0 used "T-4" for two unrelated proof obligations — a dropped thread. v2.0 cleanly partitions: T-4 = Born rule (closed in §2.7 Theorem 2.7.4); the Lorentzian-signature sufficiency is now Theorem 4.5.3 (Type-Sign sufficiency), an Established result, not a tracked target.

► M_red gap closed — Correction #5

Type: Fixed | v1.0: §9.2 T-2 row ("shared hypothesis of Thm 4.2.2 and Thm 0.6.3") | v2.0: §7.2 Table 7.2.1 T-2 row + §1.4.3

v1.0: "shared hypothesis of Thm 4.2.2 and Thm 0.6.3"

v2.0: "references Theorem 1.7.4 (thin = full) and Theorem 2.5.1 (mass-burden identity)"

Impact. Both v1.0 referent theorems were unresolved ghosts. v2.0 points T-2's blocking dependency at the actual consumers (thin=full equivalence in §1.7.4 and mass-burden identity in §2.5.1). M_red is now defined at §1.4.3 (Tier 2 sub-collective), so the v1.0 "M_red gap" — where downstream theorems referenced an undefined object — is closed.

► Acyclic dependency graph promoted from assertion to verified theorem

Type: Upgraded | v1.0: §9.5.3 (Figure 9.5.1 + narration: "Acyclicity verification — PASS") | v2.0: §7.5 Theorem 7.5.1 (Acyclic Dependency Graph) + Table 7.5.2 (Forward-Reference Resolution)

v1.0: “The cumulative Sections 0–9 dependency graph is ACYCLIC. ... No circularity exists.”

v2.0: “The cumulative dependency graph across Sections 0–6 is acyclic. ... The v1.0 asserted acyclicity but did not formally verify. The rewrite provides the verification.”

Impact. v1.0 listed forward references inline as narration (items a–i). v2.0 promotes this to a formally stated Theorem 7.5.1 with an itemized verification (§0 → §1 → ... → §7) and a structured Table 7.5.2 enumerating 11 forward-reference resolutions with statuses (RESOLVED / CERTIFIED / PARTIALLY RESOLVED). The $\delta\hat{C}$ loop fix is explicitly verified.

► λ_R honestly flagged as new free parameter

Type: Added | v1.0: §9.4.1 (λ_R listed as “Framework parameter (derived from MOE descent)”) | v2.0: §7.4 Remark 7.4.4

v1.0: “(v1.0 hid λ_R inside the S_eff component table without acknowledging it as a free parameter)”

v2.0: “ λ_R is a new free parameter introduced by the S_eff port. ... its numerical value is not derived.”

Impact. v1.0 hid λ_R inside the S_eff component table without acknowledging it as a free parameter. v2.0 explicitly flags it: S_eff introduces one new parameter that must be fixed by additional arguments or empirical input. This is part of the broader v2.0 honesty campaign (also seen in Geodesic Principle and Q-9 downgrades).

► Layer scheme reconciled — A/B/C retired

Type: Modified | v1.0: §9.5.4 Table 9.5.3 (Layer Assignment Audit, Four-Layer A/B/C/Q') | v2.0: §7.5 Remark 7.5.3 (Layer Scheme Reconciliation, $L \rightarrow Q \rightarrow C \rightarrow Q'$ only)

v1.0: “Layer A (full $\%_{kin}$, quantum-universal) ... Layer B (exact sector metric) ... Layer C (coarse-grained GR geometry)”

v2.0: “The v1.0 ‘Three-Layer Protocol’ (A/B/C) is retired. The framework uses only $L \rightarrow Q \rightarrow C \rightarrow Q'$.”

Impact. v1.0 conflated two layer schemes: $L \rightarrow Q \rightarrow C \rightarrow Q'$ (emergence ladder) and A/B/C (Three-Layer Protocol). v2.0 retains only $L \rightarrow Q \rightarrow C \rightarrow Q'$, subsuming v1.0 Layer A/B into Cubic layer C, and renaming v1.0 Layer C as Quartic layer Q' . Anti-Pattern 4 is correspondingly reworded.

Section Impact Assessment

The sign/factor-of-2 fix in Theorem 7.4.2 is the most consequential mathematical correction. v1.0's claim that $G_{\mu\nu} = \kappa_B \cdot T^A B_{\mu\nu}$ with $T^A B_{\mu\nu} = \delta/\delta g^{\mu\nu} \text{Tr}(\rho F_E)$ was simply wrong: the variation of $(1/2\kappa_B)R$ produces $(1/2\kappa_B)G_{\mu\nu}$ (not $G_{\mu\nu}/\kappa_B$), and the variation of $-\text{Tr}(\rho F_E)$ produces $-\delta/\delta g^{\mu\nu} \text{Tr}(\rho F_E)$, so v1.0's stated equation was off by a factor of 2 (and the implicit sign of the matter term was inconsistent with GR conventions). By adopting the standard GR stress-energy convention $\Theta^A(B)_{\mu\nu} := (2/\sqrt{g}) \delta/\delta g^{\mu\nu} \text{Tr}(\rho F_E)$, v2.0 makes the variation yield $G_{\mu\nu} = \kappa_B \cdot \Theta^A(B)_{\mu\nu}$ cleanly. This is the framework's central GR-recovery theorem, and its correction is essential for the $\Lambda_B = 0$ derivation, the Newtonian limit, and the cosmological Friedmann program in §6 to hang together — without the fix, all downstream GR claims inherit a factor-of-2 inconsistency.

The T-4 dual-use resolution closes a genuine dropped thread: v1.0 used “T-4” to mean both the Born rule (an EXISTENTIAL closed theorem-target) and the Lorentzian-signature sufficiency (an open structural condition referenced in three different sections). A reader of v1.0 §9.2 encountered “T-4 CLOSED” but could not reconcile this with “T-4” appearing as an open condition in §§1.7.4, 2.8.2, 5.4.1. v2.0 cleanly partitions: T-4 =

Born rule only (closed in §2.7 Theorem 2.7.4); the Lorentzian-signature sufficiency is now Theorem 4.5.3, an Established result of Type-Sign Coupling. The M_{red} gap closure complements this: v1.0 §9.2's T-2 row referenced "Thm 4.2.2 and Thm 0.6.3" — neither of which actually resolved — leaving T-2's blocking dependency on theorems that did not exist. v2.0 references the real consumers (Theorem 1.7.4 thin=full; Theorem 2.5.1 mass-burden identity) and defines M_{red} explicitly at §1.4.3 (Tier 2 sub-collective, separation of M).

Overall manuscript closure status: Phase B is complete and acyclic (Theorem 7.5.1 verified, not merely asserted). Of the seven theorem targets, T-3 is STRENGTHENED, T-4 is CLOSED, T-6 is PARTIAL, and T-1/T-2/T-5/T-7 remain OPEN — with T-2 explicitly identified as the EXISTENTIAL central open problem whose closure would convert the framework from "architecturally sound, mathematically incomplete" to "architecturally sound, mathematically established" for the majority of its conditional claims. Phase C (integration into a single ~150–180 page canonical manuscript with three-pass review) is pending.

Key Quotes from v2.0 §7

The following verbatim quotes from v2.0 best capture the section's self-described changes from v1.0:

"This section is the v1.0 Section 9 largely intact in architecture, with six corrections."

"The v1.0 stated $G_{\mu\nu} = \kappa_B \cdot T^{\mu\nu}$, but the S_{eff} variation gives $G_{\mu\nu} = -2\kappa_B \cdot T^{\mu\nu}$."

"The v1.0 §9.1 had 6 of 9 Section 5 rows pointing to wrong section/theorem numbers."

"The v1.0 claimed '13 PARTIAL' but listed 11 items. The rewrite uses '11 PARTIAL' consistently."

"T-4 is reserved exclusively for the Born rule (closed in §2.7). No more dropped thread."

"The v1.0 §9.2 T-2 row referenced 'Thm 4.2.2 and Thm 0.6.3' — neither of which resolved."

"Q-9 is not CLOSED because T-2 is open. ... corrected to 'CONDITIONAL (on T-2).'"

"The v1.0 'Three-Layer Protocol' (A/B/C) is retired. The framework uses only $L \rightarrow Q \rightarrow C \rightarrow Q'$."

Chapter 5 — Aggregate Impact Assessment

The v2.0 rewrite is, in the framework's own language, a transition from "architecturally sound, mathematically incomplete" (v1.0) to "architecturally sound, mathematically established for the majority of its conditional claims" (v2.0, conditional on T-2 closure). The rewrite does not add new physics beyond the dark energy mechanism at §2.2 — it reorganizes, fixes, and honestly downgrades. The net effect on the framework's epistemic posture is substantial.

5.1 What v2.0 Establishes That v1.0 Did Not

v2.0 establishes three things v1.0 did not. First, the acyclic dependency graph is now a formally verified Theorem 7.5.1 rather than a narrated assertion. Every section closes with an Acyclicity Test that itemizes each object's dependencies by section number and issues a verdict; the cumulative graph verification is structured in Table 7.5.2 with 11 forward-reference resolutions, each tagged RESOLVED, CERTIFIED, or PARTIALLY RESOLVED. The forward-reference contracts and $\delta\hat{C}$ loop fix that v1.0 had used as defensive patches are no longer needed because the circularities have been structurally eliminated. Second, M_{red} is now formally defined at §1.4.3 as the Tier 2 sub-collective of M — framed as "separation, not reduction." This closes a quiet but consequential gap: v1.0 had referenced M_{red} in §4.1.2 (field admissibility), §4.2.7 (Particle Identity), and §9.2's T-2 row, but never defined it. The downstream effect is that the Born rule (Thm 2.7.4) now has a genuine objectivity premise (from M_{red} 's sectoral structure) rather than the vacuous premise it inherited from v1.0's tautological Thm 7.4. Third, the dark energy mechanism at §2.2 is genuinely new physics. v1.0 §8.3.3 had treated dark energy only qualitatively as "the dynamic pressure of open extension at the causal frontier"; v2.0 §2.2 introduces restorative power $\mathbb{P}_{\text{restore}} = \eta \cdot \|\nabla_{\{\text{Bures}\}} B_{\Delta}\|^2$, the burden-dependent relaxation rate $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$, and the dark-energy density scaling $\rho_{\text{DE}} \sim (H/\Gamma)^2 \cdot \rho_{\text{source}}$. The 120-order suppression falls out as $(H_0 \cdot t_P)^2 \sim 10^{-122}$ because the exponent 2 is structural (from $B_{\Delta} = \|\text{mismatch}\|^2$), not tuned. The cosmological application in §6.3 derives four downstream results: the 120-order resolution, the $(1+z)^{-6}$ early-dark-energy suppression (resolving the CMB $\Omega_{\text{EDE}} < 0.004$ bound), an evolving equation of state $w(t)$ with three falsifiable predictions against DESI/Euclid, and the asymptotic decay $\rho_{\text{DE}} \rightarrow 0$.

5.2 What v2.0 Honestly Downgrades

v2.0 honestly downgrades nine v1.0 status claims. The most consequential: the D=3 closure "Theorem" (v1 §2.7.3) is demoted to "[Conditional Proposition]" because the closure defect $\Xi_{\text{C}}(D)$ had been defined tautologically as $|D-3|$; the Einstein closure "Strengthened" (v1 §5.4.1 P5) is reverted to "Conditional" because the smooth 4D Lorentz-compatible continuum-limit premise (T-2) is open; the Bekenstein-Hawking "T-6 partially closed" (v1 §6.5.1) is downgraded to "T-6 area-scaling only; coefficient open" because v1.0 had assumed the exponential microstate growth that was its conclusion; $\Lambda_{\text{B}} = 0$ "Theorem 8.5.1" is downgraded to "Corollary 6.4.1" because it was conditional on T-2 and a theorem conditional on an unproved conjecture is not a theorem; and the Q-9 "CLOSED (conditional on T-2)" status is reworded to "CONDITIONAL (on T-2)" because "CLOSED" and "conditional on an open target" cannot both hold. Two v1.0 theorems in §6 (Lowest-Burden Emission, Hawking-Like Emission) are demoted to a Principle and a Conjecture respectively, because their proofs "restated the conclusion." The v1.0 Theorem 5.1 (Symmetry of Burden Tensor) was tautological — "if Π is symmetric then Θ is symmetric" — and v2.0 derives $\Theta_{\mu\nu} = \Theta_{\nu\mu}$ from Hessian

symmetry. The v1.0 Theorem 4.1 (Particle Identity) had its hypotheses as its conclusion; v2.0 explicitly labels it a definitional equivalence. The v1.0 Thm 5.2 (Active-Source Conservation) assumed the field equation it was supposed to derive; v2.0 derives conservation independently via diffeomorphism invariance of B_Δ . Net effect: the framework claims less in v2.0 than in v1.0, but what it claims, it claims honestly. The status taxonomy is now strictly enforced: "Established" requires a complete proof; "Conditional" requires an open theorem-target; "Conjecture" requires an explicit derivation path; "Principle" is reserved for variational assumptions that are not derivations; "Corollary" is reserved for results conditional on an Established theorem.

5.3 What v2.0 Leaves Unchanged (Mathematically)

The mathematical content of v2.0 is largely v1.0's, with the same algebraic objects (A, M, F, R_t), the same propagator composition $R_t = \lim_{N \rightarrow \infty} (R_{\{T/N\}^{\{OExt\}}} \circ (R_{\{e\}^{\{OExp\}}})^N)$ (only the SOE/MOE labels renamed), the same convergence theorem and Master-Zero equivalence, the same thin/full A_{phy} split, the same Dirac bracket formula, the same correlation kernels ($K_\omega, K_\omega, K_\omega(A,B,C)$), the same logarithmic distance $d_\omega = -\ell_0 \log K_\omega$, the same burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$, the same burden tensor $\Theta^\wedge(B)_{\mu\nu}$ as variational derivative of B_Δ , the same ADM dictionary, the same metric boundedness $\det(h) \geq \ell_0^{2d}$, the same dual-horizon ratios (χ_R, ε_R), the same boundary-record map, and the same nine interpretive guardrails. What changes is the location, framing, and epistemic status of these objects. R_t moves from §0.4 to §1.5 (after the fracture machinery that licenses it). The convergence theorem moves from §0.5 to §1.6. The thin/full split moves from §0.6–0.7 to §1.7. The Dirac bracket moves from §0.7b to §1.4.5. The Reconciliation Principle moves from §0.8 to §2.1 (after burden has its physical meaning). Π_{net} moves from §0.3.8 to §3.9 (after K_ω is available). Burden moves from §3 to §1.3 (where it receives its physical interpretation). Fields, particles, mass, records, Born, and FIREWALL move from §4 and §7 into the new §2 (the QM layer). Time and gravity move from §3 and §5 into the new §4 (the GR perspective). The mathematical content is preserved; the architectural coherence is repaired.

Chapter 6 — Open Issues Carried Forward

v2.0 closes some open issues from v1.0 (the M_{red} gap, the T-4 dual-use bug, the Q-9 status contradiction, the Π_{net} forward reference, the dark energy mechanism linkage to §2.2, the D=3 circularity, the Einstein-closure conservation circularity, the Arrow-of-Time generator swap, the Newtonian sign error, the BH coefficient circularity, the missing $\Rightarrow$ in Eq. 6.4.3, the layer-badge inconsistencies, and seven stale cross-references in §6). It does not close the four open theorem targets T-1, T-2, T-5, and T-7. Of these, T-2 is explicitly identified in §7.2 as the existential central open problem whose closure would convert the framework from "architecturally sound, mathematically incomplete" to "architecturally sound, mathematically established" for the majority of its conditional claims.

6.1 The Four Open Theorem Targets

Target	Statement (v2.0)	Blocking	Priority
T-1	Derivation of γ (Open Expansion rate) from spectral gap of F on $\ker(M)$.	Spectral analysis of F not yet performed.	#2
T-2	Continuum limit: stable-mode convergence to smooth 4D Lorentz-compatible manifold.	Smooth-manifold hypothesis unproven; h_{eff} numerical coefficient undetermined.	#1 (EXISTENTIAL)
T-5	Cubic volume $E_\omega \leftrightarrow$ scale factor $a(t)$ connection at cosmological scale.	Non-local relational-channel $\Theta^{(rel)}$ FLRW averaging unresolved.	#4
T-7	Poisson recovery from relational burden (quantitative).	Quantitative match of $\Delta\Phi_{kk}$ gravitational potential profile.	#3

Why T-2 Is Existential

T-2 underlies the Convergence Theorem (Thm 1.6.1), the thin=full equivalence (Thm 1.7.4), the mass-burden identity (Thm 2.5.1), the Dirac bracket (Thm 1.4.6), $\Lambda_B = 0$ (Cor 4.5.5), the Einstein closure (Thm 4.5.4), the Friedmann equation (Conj 6.2.2), the BH area scaling (Thm 5.5.3), and the dark energy mechanism's amplitude (Conj 6.3.2). Closing T-2 in one stroke would convert most of the framework's Conditional results to Established. Until T-2 is closed, the framework remains architecturally sound but mathematically incomplete.

6.2 Phase C Roadmap

v2.0 §7.7 announces Phase C as the next stage: integration of the eight section PDFs into a single ~150–180 page canonical manuscript, followed by a three-pass review (mathematical pass for proof correctness, structural pass for emergence-chain coherence, editorial pass for consistency). Phase C will carry forward the four open theorem targets listed above, plus the partial results T-3 (D=3 closure — strengthened but not fully derived) and T-6 (BH area scaling established, coefficient open). The 17-point quarantine list is preserved

verbatim from v1.0 per Spec Ch. 12.4 anti-pattern (1); the v2.0 corrections to that list's arithmetic ("11 PARTIAL" rather than v1.0's self-contradictory "13 PARTIAL") and Q-9 status ("CONDITIONAL (on T-2)" rather than v1.0's self-contradictory "CLOSED (conditional on T-2)") will be incorporated.

The framework's foundational claim — that physics can be derived from a relational algebra equipped with a master constraint and a fracture operator — remains architecturally sound in v2.0. The rewrite does not challenge the claim; it makes the architecture checkable, the proofs honest, and the open problems explicit. Phase C is where the open problems get closed, one by one, starting with T-2.